ROYAL CANADIAN NAVY QUALIFICATION STANDARD AND PLAN (QSP)

Marine Technician (MAR TECH)

USQ – Engineering Watchkeeper ORCA
(NQual AJLQ)

MITE Course IDs:

Residential: 116876
OJPR: 117573

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RECORD OF AMENDMENTS

1. This QSP is developed under the authority of the Royal Canadian Navy (RCN) through the Training Authority (TA), Naval Personnel and Training Group (NPTG) and the Commandant of the Curriculum Control Authority CCA (NTDC(P)). The approval or revision date should appear in the footer of every page as an additional means of verification. Details on the change(s) must be amplified in Annex J "Course History".

Revision	Approval Date	Authority
1.0	March 2020	NPTG
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FOREWORD

1. This training document is developed under the authority of the Commander Naval Training and Personnel Group (CNPTG) and in accordance with the following references:

   a. A-P9 – 050 Series, Canadian Forces Manual of Individual Training and Education (CFITES);

   b. Annex C – 5555-00379-MAR-TECH dated 15 Dec 16;

   c. NAVORD 4500-0 - Structured Delegation; and

   d. NPTORD 4500-22 – Qualification Standard and Plan Production and Management.

2. This QSP consists of specific Performance Objectives (POs) in Chapter 2 and Lesson Specifications and Enabling Objectives (EOs) in Chapter 4. A trainee will achieve the qualification when the standards listed in the POs have been met.

3. This QSP is effective upon approval and supersedes previous versions.

4. Suggestions for amendments to this QSP may be initiated by anyone and shall be forwarded in writing to the CCA, who will forward to NPTG for approval, as appropriate.

5. The designated TE(s) for this training is indicated below:

TE	Phase	MITE Course ID		NQual
		Residential	OJPR	
NFS(P)	All	116876	117573	AJLQ

CHAPTER 1: TRAINING MANAGEMENT DETAILS

AUTHORIZATION FOR THIS CHAPTER AND AMENDMENTS MUST BE GRANTED BY
COMMANDER NAVAL PERSONNEL AND TRAINING GROUP (CNPTG)

AIM

1. The aim of this Qualification Standard and Plan (QSP) is to establish the requirements and training for the ORCA Class Engineering Watch keeper. This QSP is designed to assist training establishments in developing, conducting, evaluating and validating the training.

COURSE DURATION

2. Course duration is calculated in accordance with NPTORD 4500-22- Qualification Standard and Plan, para 4.5. For this course, a training day consists of 400 minutes.

	Minutes
Actual Training Time (ATT) (Sum of all EOs/POs excluding PC&ECs)	6950
Enabling Checks (ECs)	750
Performance Checks (PCs)	6700
Sub-total	14400 min 36 **trg days**
Allocation of BAT, TAT, TTT, PT IAW NPTORD 4500-22, Table 5.5	
Base Administrative Time (BAT)	400
Training Administrative Time (TAT)	200
Training Travel Time (TTT)**	200
Physical Fitness (PT)	800
Sub-total	1600 min 4 **trg days**
Course Duration Calculation (ATT+PCs&ECs+BAT+TTT+TAT+PT) * **Partial days must be rounded up** ** **TTT is intended for travel to and from PT, 50 minutes per PT period**	16000 min 40 **trg days**

RESOURCES

3. Identify key personnel and facilities required to conduct the training. Additional training establishment resources can be listed in Annex H if required.

Required Course Staffing					
EO	EO Name	Min Qualification	Min Rank	Qty for Instruction	Qty for Assessment
001.01	Explain the IPCS	Mar Tech – MS/PO2 OCE OR MAR TECH ORCA Wkpr qualified.	MS/ PO2 MS/ PO2	1	1
001.02	Explain the Ancillary, Auxiliary and Domestic Systems				
001.03	Explain the Power Generation and Distribution System				
001.04	Explain the Shaftline Arrangements				
001.05	Explain Main Gearing				
001.06	Explain Propulsion Diesel Engine				
002.01	Operate Bridge Control Console				
002.02	Operate Auxiliary Systems				
002.03	Operate Ancillary Systems				
002.04	Operate Domestic Systems				
002.04	Operate Power Generation and Distribution (PG&D)				
002.05	Operate Gearing & Shafting				
002.06	Operate Main Propulsion				
003.01	Investigate Auxiliary Systems				
003.02	Investigate Ancillary Systems				
003.03	Investigate PG&D				
003.04	Investigate Gearing & Shafting				
003.05	Investigate Main Propulsion				
004.01	Respond to Auxiliary and Ancillary Alarms				
004.02	Respond to PG&D Alarms				
004.03	Respond to Main Propulsion and Gearing and Shafting Alarms				
004.04	Respond to Ship’s Emergencies				
005.01	Complete Fueling Orders				
005.02	Lead Maritime Vessel Fueling				
OJPR	On-Board Phase	Sr. Eng	PO1/ PO2	Nil	1

Critical Training Resources	No of Days
Operational Orca Class Vessel	30
NFS(P) Classroom	10

It is possible a Fleet School may not have the necessary equipment or other resources to conduct training as laid out in this QSP. In such cases, alternate methods as documented by Training Deviation Requests (TDRs) may be considered and/or changes in scheduled timings for the course shall be considered.

COURSE CAPACITY

4. Planning for the conduct of training is based on the guidelines in NPTORD 4500-40 – Quantity Control Process for the NTS. Resource implications need to be considered if a course size is above the maximum load.

    a. Maximum: 8;

    b. Minimum: 4; and

    c. Optimum: 6.

MITE

5. Loading. The authority to load members onto this course lies with the Training Establishment (TE). NPTG is to be included as an INFO addressee on all loading messages.

6. Scheduling. Each session of this course shall be scheduled IAW NPTORD 4500-1 – Individual Training and Education Policy and NPTORD 4500-40 Quantity Control Management for the Naval Training System; in MITE by the Training Establishment conducting the training, and NPTG through the Quantity Control section. The course length in MITE is to be based on the course duration, planned annual leave, training breaks, weekends, and training system holidays. Course sessions for Naval Reserve Force may be scheduled during training system holidays.

7. Prerequisites. To be selected for this course the loading authority must ensure that trainees have the following prerequisites:

    a. One of the following:

        i. USQ – Roundsperson HFX (MITE 121178/121179, NQual ALMJ); or
        ii. USQ – Roundsperson KIN (MITE 121180/121181, NQual ALMK); or
        iii. USQ – Roundsperson ORCA (MITE 121182/121183, NQual ALML); and

b. One of the following:
   i. USQ S1 – Mechanical (MITE 122019, NQual TBD); or
   ii. QL5 MAR ENG (MITE 109363/109364, NQual AKTO); or
   iii. QL5 HTECH (MITE 108493, NQual AIFF); or
   iv. QL5 ETECH (MITE 109328/109329, NQual AJTL); or
   v. KINGSTON Class B Ticket (NQual AJAB).

8. Qualification. Successful completion of this course constitutes the primary means of obtaining the Engineering Watchkeeper ORCA Class (NQual: TBD).

COURSE REPORTING

9. Upon passing this course, trainees shall receive a Course Report (CF 377), Certificate of Military Achievement (CF 289) completed IAW CMP Instruction 01/06 - Course Reporting and Certificates and NPTORD 4500-51 - Course Reports and Certification. Monitor Mass Training Tracker will be updated to reflect the members results.

EVALUATION

10. Program evaluation shall be conducted IAW the guidance provided in A-P9-050-000 CFITES Vol. 11 - Evaluation and Validation Techniques.

VALIDATION

11. This training will be validated every five years CFITES Vol. 8 - Validation of Instructional Programs and based on the task list in Annex B.

PRIOR LEARNING ASSESSMENT AND RECOGNITION (PLAR)

12. PLARs shall be conducted in accordance with NPTORD 4500-27 – PLAR, DAOD 5031-1 - Canadian Forces Military Equivalencies Program and CFITES Vol 12 - Canadian Forces Military Equivalencies Program: Prior Learning Assessment.

SPECIAL INSTRUCTIONS

13. The following are pre-requisites for a member to be loaded on the USQ – Engineering Watchkeeper ORCA course:

    a. One of the following:

(1) USQ – Roundsperson HFX (MITE 121178/121179, NQual ALMJ); or

(2) USQ – Roundsperson KIN (MITE 121180/121181, NQual ALMK); or

(3) USQ – Roundsperson ORC (MITE 121182/121183, NQual ALML); and

b. One of the following:

(1) USQ S1 – Mechanical (MITE 122019, NQual TBD); or

(2) QL5 MAR ENG (MITE 109363/109364, NQual AKTO); or

(3) QL5 HTECH (MITE 108493, NQual AIFF); or

(4) QL5 ETECH (MITE 109328/109329, NQual AJTL).

CHAPTER 2: PERFORMANCE OBJECTIVES

AUTHORIZATION FOR THIS CHAPTER AND AMENDMENTS MUST BE GRANTED BY
COMMANDER NAVAL PERSONNEL AND TRAINING GROUP (CNPTG)

PO 001 EXPLAIN THE OPERATION OF ENGINEERING SYSTEMS

1. Performance. Explain the operation of engineering systems.

2. Conditions.

   a. Given:

      (1) Operational IPCS;

      (2) References; and

      (3) Supervision.

   b. Denied: Assistance.

   c. Environmental: At sea or alongside in any weather condition.

3. Standard. The Engineering Watchkeeper shall, IAW references, explain the operation of engineering systems including:

   a. IPCS;

   b. Ancillary, Auxiliary and Domestic systems;

   c. Power Generation & Distribution (PG&D) system;

   d. Shafting arrangements;

   e. Main Gearing; and

   f. Main Engine (ME).

4. References. A1, A2, A3, A4, A5, A6, A7, C2, C3, C4, C5, C6, C7, C8, C9, C10, C11, C12, C13, C14, C15, C16, C19, C20, C21, C22, C23, C24, C31, C32, C33, C34, C35, C37 C38, C39, C40, C42, C43, C44, C45, C57, C58, C59.

5. Task Numbers. AUX001, AUX002, AUX007, AUX008, AUX014, AUX015, AUX026, AUX029, AUX049, AUX050, AUX053, 060, AUX066, AUX067, AUX074, AUX080, AUX086, AUX141, AUX170, AUX171, PGD002, PGD003, PGD004, PGD021, PGD022, PGD025, PGD027, FDC001, FDC002, FDC003, FDC004, FDC012, FDC013, WKP006, WKP007, WKP008, WKP013, WKP023, WKP024, WKP026, WKP028, WKP029, WKP031, WKP032, WKP033, WKP034, WKP035, WKP036, WKP042, WKP043, WKP044, WKP045, WKP046, WKP047, WKP050, WKP052, WKP061, WKP064, WKP065, WKP066, WKP067, WKP068, WKP069, WKP070, WKP071, WKP072, WKP073, WKP074, WKP075, WKP076, WKP077, WKP078, WKP079, WKP083,

WKP084, WKP088, WKP089, WKP090, WKP091, WKP093, WKP095, WKP099, WKP100, WKP101, WKP102, WKP103, WKP105, WKP106.

6. Training Limitations. Nil.

PO 002 OPERATE INTEGRATED PLATFORM CONTROL SYSTEM (IPCS) CONSOLE

1. Performance. Operate IPCS system.

2. Conditions.

   a. Given:

      (1) Ship with applicable equipment and/or system;

      (2) PPE;

      (3) Operational IPCS system;

      (4) Assistance;

      (5) References; and

      (6) Supervision.

   b. Denied: Nil.

   c. Environmental: Any conditions, day or night, at sea or alongside.

3. Standard. The Engineering Watchkeeper shall operate IPCS system IAW refs by:

   a. Operating Bridge Control console;

   b. Operating auxiliary systems;

   c. Leading Collective Operation of ancillary systems;

   d. Operate Domestic Systems;

   e. Operating PG&D system;

   f. Operating gearing and shafting systems;

   g. Operating main propulsion system; and

4. References. A1, A2, A3, A4, A5, A6, A7, C2, C3, C4, C5, C6, C7, C8, C9, C10, C11, C12, C13, C14, C15, C16, C19, C20, C21, C22, C23, C24, C31, C32, C33, C34, C35, C37 C38, C39, C40, C42, C43, C44, C45, C57, C58, C59.

5. Task Numbers. CT0350, WT0010, IT0380, ALT1520, HT0060, HT0040, HT0050, HT0070, ET0230, DT0120, ATT0920, XT0090, DT0070, DT0060, CT0060, CT0280, ABT0050, BT0200, ABT0150, ABT0170, RT0010, ABT0200, QT0390, ABT0210,

AKT1090, GT0010, QT0350, ABT0070, ABT0080, CT0270, WT0040, WT0090, IT0510, AMT0160.

6. Training Limitations. NIL.

PO 003 INVESTIGATE ENGINEERING SYSTEMS

1. Performance. Investigate engineering systems.

2. Conditions.

   a. Given:

      (1) Ship with applicable equipment and/or system;

      (2) PPE;

      (3) Operational IPCS;

      (4) Assistance;

      (5) References; and

      (6) Supervision.

   b. Denied: Nil.

   c. Environmental: Any conditions, day or night, at sea or alongside.

3. Standard. The Engineering Watchkeeper shall investigate engineering systems IAW REFs by:

   a. Investigating collective operation of auxiliary systems;

   b. Investigating collective operation of ancillary systems;

   c. Investigating PG&D system;

   d. Investigating operation of gearing and shafting systems;

   e. Investigating main propulsion system; and

   f. Investigating collective operation of domestic systems.

4. References. A1, A2, A3, A4, A5, A6, A7, C2, C3, C4, C5, C6, C7, C8, C9, C10, C11, C12, C13, C14, C15, C16, C19, C20, C21, C22, C23, C24, C31, C32, C33, C34, C35, C37 C38, C39, C40, C42, C43, C44, C45, C57, C58, C59.

5. Task Numbers. AHT0860, AHT0840, CT0130, AVT1310, AVT1340, AVT1280, AVT1300, AVT1290, AVT1360, AVT1330, AVT1320, AVT1350, AHT0850, AHT0880, BT0050, BT0060, CT0120, AHT0820, CT0160, CT0150, CT0430, CT0140.

6. Training Limitations. Nil.

PO 004 RESPOND TO ALARMS AND EMERGENCIES

1. Performance. Respond to alarms and emergencies.

2. Conditions.

   a. Given:

      (1) Ship with applicable equipment and/or system;

      (2) PPE;

      (3) Operational IPCS;

      (4) Assistance;

      (5) References; and

      (6) Supervision.

   b. Denied: Nil.

   c. Environmental: Any conditions, day or night, at sea or alongside.

3. Standard. The Engineering Watchkeeper shall respond to alarms and emergencies IAW REFs by:

   a. Responding to auxiliary, ancillary and domestic systems alarms;

   b. Responding to PG&D alarms;

   c. Responding to main propulsion and gearing and shafting systems alarms; and

   d. Responding to ship's engineering emergencies.

4. References. A1, A2, A3, A4, A5, A6, A7, C2, C3, C4, C5, C6, C7, C8, C9, C10, C11, C12, C13, C14, C15, C16, C19, C20, C21, C22, C23, C24, C31, C32, C33, C34, C35, C37 C38, C39, C40, C42, C43, C44, C45, C57, C58, C59.

5. Task Numbers. JT0040, CT0330, CT0050, JT0030, CT0030, PT0010, CT0340, CT0310, CT0320, CT0010, CT0020, CT0040, GT0010, WT0020, IT0470.

6. Training Limitations. Nil.

PO 005 Perform the Duties of Fueling I/C

1. Performance. Perform the Duties of Fueling I/C.

2. Conditions.

   a. Given:

      (1) Operational ship with fueling capabilities;

      (2) Qualified fueling party;

      (3) References;

      (4) Communications equipment;

      (5) Supervision;

      (6) Fuel spill response kits;

      (7) HAZMAT containment capabilities;

      (8) Personal PPE;

      (9) Fuel; and

      (10) Updated fuel report.

   b. Denied: Nil.

   c. Environmental: Onboard ship, alongside, in any condition.

3. Standard. The Marine Technician Mechanical Supervisor, adhering to all safety precautions, shall independently perform the duties of the fueling I/C IAW refs by:

   a. Initiating fueling orders;

   b. Leading maritime vessel fueling (internal fuel/ consolidation) (OT0090) to include:

      (1) Preparing a ship to move (embark /disembark / transfer) fuel;

      (2) Moving (embark / disembark / transfer) fuel; and

      (3) Securing a ship from fueling operations.

   c. Finalizing fueling orders

4. References.

5. Task, Skill, Knowledge Numbers. OT0090, OT0010.

6. Training Limitations. To effectively instruct the TE requires an operational system control interface (IPMS/IPCS).

CHAPTER 3: ASSESSMENT OF TRAINEE

AUTHORIZATION FOR THIS CHAPTER AND AMENDMENTS MUST BE GRANTED BY NTDC(P)

GENERAL

1. All Performance Checks (PCs) are based on Performance Objectives (POs) in Chapter 2 of this QSP and the standard listed must be met to achieve this qualification. PCs will be administered in the form of practical examinations and the trainee must obtain a pass in all POs to successfully complete this course.

ENABLING AND PERFORMANCE CHECKS

2. Enabling Checks (ECs). ECs are formative tests that help the trainee and the instructors recognize progress or gaps in learning, so that confirmation or corrective action (such as additional practice or remedial instruction) can be provided. They also reinforce learning, so that it will be better retained for longer periods. Formative tests are designed to give feedback to both the instructor(s) and trainees. Marks for ECs are not calculated in the trainee's final mark.

3. Performance Checks (PCs). PCs are summative tests used to determine if the trainee has met the standard under the conditions (where possible) of the PO and hence can do the job. Supplemental PCs are administered upon failure of a PC and must be in the same format and have the same standard as the original PC.

RESPONSIBILITY FOR TESTING

4. Assessment of trainees is to be done IAW CFITES Vol. 7 – Evaluation of Learners.

5. Senior instructor or delegated staff provides quality control by administering and scoring all Performance Checks. Any PC not administered or invigilated by staff will be administered by a non-involved instructor designated by the senior instructor. Enabling Checks are the responsibility of the Instructional Staff.

PROGRESS MONITORING

6. Monitoring learning is the responsibility of all instructional staff and instructional supervisors. Monitoring, including unsatisfactory progress, is done IAW CFITES Vol 6 - Conduct of Instruction Programs and TE SOs or SOPs.

7. Continuous monitoring of candidate progress is required on this course in order to provide the following:

   a. Early warning of trainee difficulties;

   b. Feedback on the effectiveness of training;

   c. Information for the TRB or PRB (if required); and

d. Information to assist in completion of the CF 377 Course Report.

8. Trainees experiencing difficulty in any area of performance are to be counselled WRT the nature of their shortcomings and suggested corrective action IAW Training Establishment SOs or SOPs.

9. PC failures and supplemental PC failures are to be handled IAW Training Establishment SOs or SOPs. Supplemental PCs shall not be given prior to 24 hours having elapsed since trainee was informed of their PC failure.

TRAINING REVIEW BOARD

10. TRBs shall be convened and conducted IAW with NPTORD 4500-54 - Training Review Board Process.

COURSE REPORTS

11. Course reports (CF 377) shall be prepared and distributed IAW CFAO 26-12, NPTORD 4500-51 - Course Reporting and Certificates.

12. Trainees will be graded pass or fail. IAW NPTORD 4500-51 - Course Reporting and Certificates, students will not be ranked.

13. Where necessary, the narrative will indicate trainee's strengths and weaknesses based on PC results. Training limitations and/or incomplete training must also be included in the course report. Record of trainee counselling is to be maintained by the training establishment.

14. When training is completed Monitor Mass Training Tracker shall be updated to reflect the results of the course and new qualification, if applicable.

CHAPTER 4: LESSON SPECIFICATIONS

AUTHORIZATION FOR THIS CHAPTER AND AMENDMENTS MUST BE GRANTED BY NTDC(P)

PO 001 Explain the Operation of Engineering Systems

EO 001.01 Explain the IPCS.

1. Performance. Explain the IPCS.

2. Conditions.

    a. Given:

        (1) Operational IPCS/Orca Class Vessel;

        (2) Schematics and illustrations from naval and manufacturers' manuals; and

        (3) References.

    b. Denied: Nil.

    c. Environmental: Orca Class Vessel Training Room.

3. Standard. The trainee shall, IAW references, explain the IPCS including:

    a. Architecture:

        (1) Network Configuration;

        (2) Human Machine Interfaces (HMIs); and

        (3) Virtual Remote Terminal Units (VRTUs).

    b. Operation:

        (1) Station In Control (SICs); and

        (2) Alarms, Warnings and Errors.

4. Time.

    a. 3.5 x 50 min periods; and

    b. EC review 1 x 50 min periods.

5. Method / Media.

    a. Lecture;

    b. Demonstration;

    c. Ship tour; and

d. OJT.

6. References. A1, A2, A4, A7, C11, C22, C23, C43, C44, C45, C57, C58, C59.

7. Training Aids.

a. Power Point presentations;

b. System drawings;

c. Orca Class Manual; and

d. Orca Class Vessel.

8. Test Details. EC 001.01 will be a homework assignment to be corrected during class time. EO 001.01 will be tested in PC 001.03.

9. Remarks. OJT not included in time calculations.

10. Teaching Points.

Teaching Point	Sub Teaching Points	Instructional Technique (Time in min)			Method	Refs	Comments
		T	D	P			
TP1							
Network Configuration: 1. Location, function, general overview; 2. Distributed Ethernet LAN (DREL);		25			Lecture/ demo	A1, C44, C45, C57, C58	
TP2							
HMIs: 1. Location, function, general overview;		25	25		Lecture/ demo	A1, C44, C45, C57, C58	
TP3							
VRTUs: 1. Location, function, general overview;		25			Lecture/ demo	A1, C44, C45, C57, C58	
TP4							
SICs: 1. General overview;		25			Lecture/ demo	A1, C44, C45,	

2. SOTs and EOTs; and					C57, C58	
TP5						
Alarms, Warnings and Errors: 1. General overview; 2. Recognition and indications; and 3. Event Viewer Management.	25			Lecture/ demo	A1, C44, C45, C57, C58	
TP6						
Tests: 1. BITE	25			Lecture/ demo	A1, C44, C45, C57, C58	

Enabling check - EC 001.01 Explain the IPMS

PART I – ADMINISTRATIVE INSTRUCTIONS

Material and Facilities Instructions

1. Classroom.

Arrangements and Procedures

2. EC 001.01 will be a written homework assessment based on content from EO 001.01 and reviewed during class time. EO 001.01 will be tested as part of PC001.03.

PART II – INSTRUCTIONS GIVEN TO TRAINEE

1. EC 001.01 will be a written homework assessment based on content from EO 001.01.

2. Read each question carefully, and all responses fully, before answering the question.

3. Written homework will be reviewed and verified in a group discussion setting.

4. Do you have any questions?

PART III – ENABLING CHECK NOTES

1. Instructor to verify homework completed prior to discussion group.

2. Trainees to ensure accuracy of answers based on group discussion led by instructor.

3. Written homework to become self-study aid for PC 001.03 and PC 004.

PART IV – ENABLING CHECK LIST

N/A.

EO 001.02 Explain Ancillary, Auxiliary and Domestic Systems.

1. Performance. Explain ancillary, auxiliary and Domestic systems.

2. Conditions.

   a. Given:

      (1) Operational IPCS/Orca Class Vessel;

      (2) Schematics and illustrations from naval and manufacturers' manuals; and

      (3) References.

   b. Denied: Nil.

   c. Environmental: Orca Class Vessel Training Room.

3. Standard. The trainee shall, IAW references, explain ancillary, auxiliary and domestic systems(ORCA Class) including:

   a. Fuel Oil Systems;

   b. Clean Lube Oil fill and transfer (CLO);

   c. Steering Gear (STG);

   d. Sea Water Cooling System (SWC);

   e. Compressed Air system;

   f. Deck Crane;

   g. Fire Detection System;

   h. CO2 Fire Suppression;

   i. Fresh Water System;

   j. Black Water/Sanitary System;

   k. Grey Water System; and

   l. HVAC System.

4. Time.

   a. 23.5 x 50 min periods; and

   b. EC review – 2 x 50 min period.

5. Method / Media.

   a. Lecture;

   b. Demonstration – time captured during lecture periods;

   c. Ship tour; and

6. References. A1, A2, A4, C7, C9, C10, C11, C13, C14, C15, C16, C31, C37, C44, C45, C48, C57, C58, C9

7. Training Aids.

   a. Power Point presentations;

   b. System drawings;

   c. Orca Class Manual; and

   d. Orca Class Vessel.

8. Test Details. EC 001.02 will be a homework assignment to be corrected during class time. EO 001.02 will be tested in PC 001.3

9. Remarks. OJT not included in time calculations.

10. Teaching Points.

Teaching Point	Sub Teaching Points	Instructional Technique (Time in min)			Method	Refs	Comments
		T	D	P			
TP1							
Fuel Oil Systems – Fuel Oil Services (FOS): 1. Location, function, general overview; 2. Tanks: a. Storage; b. Days; c. DDFP Day; d. Fuel overflow; and e. Emergency fuel. 3. Pumps: a. ME and DG Boost; and b. Priming Pumps.		50	50		Lecture/ demo	A1, A2, A7, C9, C13, C15, C16, C31, C44, C45, C57, C58, C59	

4. Filters and Strainers: a. Fuel Water Separator; and b. Suction Strainers. 5. End Users: a. MEs; b. DGs; c. Diesel generators; d. Diesel Driven Fire Pump (DDFP). 6. Modes of operation: a. Normal; and b. Abnormal.						
TP2						
Fuel Oil Systems – Fuel Oil Filling and Transfer (FOT): 1. Location, function, general overview; 2. Tanks: a. Storage; and b. Days; c. Fuel Over flow; and d. DDFP Day. 3. Pumps: a. Motor driven; and b. Hand. 4. Fuel management; 5. Parameters; 6. Liquid Level Management (LLM). 7. Fuel Overflow and Vent System.	25	25		Lecture/ demo	A1, A2, A7, C9, C13, C15, C16, C31, C44, C45, C57, C58, C59	
TP3						
Clean Lube Oil fill and transfer (CLO): 1. Location, function, general overview; 2. Tank; 3. Pumps; a. Electrical; and b. Hand.	25	25		Lecture/ demo	A1, A2, A6, C7, C9, C13, C14, C15, C44, C45,	

4. Valves; a. Relief; 5. Operation:					C57, C58, C59	
TP4						
Steering: 1. Location, function, general overview; 2. Pumps: a. Main; b. Hand. 3. Tanks; 4. Filters; 5. Valves: a. DARB; and b. Directional Control (DCVs). 6. Modes of operation: a. Normal: (1) Full Follow Up (FFU); (2) Follow Up (FU); and (3) Non Follow Up (NFU). b. Emergency: (1) ESP; (2) Single rudder; and (3) Locked. 7. Parameters: a. Alarms; b. Trips; and c. Warnings.	100	50		Lecture/ demo	A1, A2, A6, A7, C10, C11, C44, C45, C57, C58, C59	
TP5						
Sea Water Cooling System 1. Location, function, general overview; 2. End users: 3. Modes of operation: a. Normal; and b. Abnormal.	25	50		Lecture/ demo	A1, A2, A4, A7, C2, C3, C4, C7, C9, C12, C13, C14, C44,	

					C45, C57, C58, 59	
TP6						
Compressed Air system: 1. Location, function, general overview; 2. End users: a. Reducing stations; b. Air priming; c. Utility outlets; d. De-weeding; and e. Ships whistle. 3. Parameters: a. Alarms; b. Trips; and c. Warnings.	25	25		Lecture/ demo	A1, A2, A4 A7, C37, C44, C45, C48, C57, C58, C59	
TP7						
Deck Crane: 1. Crane operation: a. Breaker location: (1) Hydraulic crane; and (2) Heater. b. On/off switch location; c. Parameters; and d. Limitations. 2. Safety procedures: a. RADHAZ considerations; and b. Personnel considerations. 3. Crane manipulation : a. Launching/retrieving the RIB SR2 Equipment: (1) Hand signals for crane manipulation; and (2) Crane command communication. b. Location: (1) Beside the crane; and (2) From the RIB SR2.	50	50		Lecture/ demo	A1, A2, A4 A7, C37, C44, C45, C48, C57, C58, C59	

4. Remote operation of the deck crane to include: a. Connecting and disconnecting remote controls; and b. Limitations. 5. Manual Operation of the deck crane to include: a. Operation of deck crane SOLAS controls when vessel is without power: and 6. Limitations.						
TP8						
Fire Detection and Fire Main System. 1. Fire main system: a. Layout; (1) Equipment; and (2) Diesel driven fire pump (DDFP). b. Operation; and c. Cold weather precautions. 2. Fire Detection System a. Sensors (1) Heat; and (2) Smoke b. Pull Stations c. Control Panel d. Strobe lights and bells	100	50		Lecture/ demo	A1, A2, A7, C9, C13, C15, C16, C31, C44, C45, C57, C58, C59	
TP9						
CO2 Fire Suppression System: 1. Location, function, general overview; 8. CO2 Compartment: a. Bottles; and b. Valves; c. Pressure Switches; and	25	25		Lecture/ demo	A1, A2, A7, C9, C13, C15, C16, C31, C44, C45, C57,	

d. Local Activation. 9. Command Considerations: a. Operating Co2 System: b. Hand.					C58, C59	
TP10						
Fresh Water System: 1. Location, function, general overview; 2. Operation: a. Tanks; and b. Pumps; c. Hot fresh water; d. Embarkation; and e. Disembarkation.	50	50		Lecture/ demo	A1, A2, A7, C9, C13, C15, C16, C31, C44, C45, C57, C58, C59	
TP11						
Black Water/Sanitary System: 1. Layout; a. Pump; b. Vacuum manifold; c. Level-o-meters. d. Accumulator; and e. Strainers. 2. Operation: a. Disembarking/flushing. b. Parameters 3. Safety Environmental management system (SEMS).	50	50		Lecture/ demo	A1, A2, A7, C9, C13, C15, C16, C31, C44, C45, C57, C58, C59	
TP12						
Grey Water System: 1. Layout; a. Pump; b. Anti-siphon vacuum breaker	25	25		Lecture/ demo	A1, A2, A7, C9, C13, C15, C16, C31, C44, C45, C57, C58, C59	
TP13						

HVAC System: 1. Layout; a. Fan coil units (FCUs); b. Electrical heaters; c. Electrode Steam Humidifiers; d. Electric Duct heater; and e. Compressor. f. Wheelhouse (bridge) Air Handling Unit (AHU); g. Main AHU; and h. Engine room temperature regulator. 2. Operation.	100	50		Lecture/ demo	A1, A2, A7, C9, C13, C15, C16, C31, C44, C45, C57, C58, C59	

Enabling check - EC 001.02 Explain Ancillary and Auxiliary Systems.

PART I – ADMINISTRATIVE INSTRUCTIONS

Material and Facilities Instructions

1. Classroom.

Arrangements and Procedures

2. EC 001.002 will be a homework assessment to be reviewed during class time. EO 001.02 will be tested under PC 001.03.

PART II – INSTRUCTIONS GIVEN TO TRAINEE

1. EC 001.02 will be a written homework assessment based on content from EO 001.02.
2. Read each question carefully, and all responses fully, before answering the question.
3. Written homework will be reviewed and verified in a group discussion setting.
4. Do you have any questions?

PART III – ENABLING CHECK NOTES

1. Instructor to verify homework completed prior to discussion group.
2. Trainees to ensure accuracy of answers based on group discussion led by instructor.
3. Written homework to become self-study aid for PC 001.03 and PC 004.

PART IV – ENABLING CHECK LIST

N/A.

EO 001.03 Explain the Power Generation and Distribution System (PG&D).

1. Performance. Explain the PG&D system.

2. Conditions.

    a. Given:

        (1) Operational IPCS/Orca Class Vessel;

        (2) Schematics and illustrations from naval and manufacturers' manuals; and

        (3) References.

    b. Denied: Nil.

    c. Environmental: Orca Class Vessel Training Room.

3. Standard. The trainee shall, IAW references, explain the PG&D system including:

    a. Caterpillar diesel generator set; and

    b. Electrical distribution system.

4. Time.

    a. 5.5 x 50 min periods; and

    b. EC Review – 1 x 50 min periods.

5. Method / Media.

    a. Lecture;

    b. Demonstration;

    c. Ship tour; and

    d. OJT.

6. References. A1, A2, A4, A7, C2, C3, C6, C7, C8, C38, C39, C40 C44, C45, C57, C58, C59

7. Training Aids.

    a. Power Point presentations;

    b. System drawings;

    c. Orca Class Manual; and

d. Orca Class Vessel.

8. Test Details. EC 001.03 will be a homework assignment to be corrected during class time. EO 001.03 will be tested in PC 001.03

9. Remarks. OJT not included in time calculations.

10. Teaching Points.

Teaching Point	Sub Teaching Points	Instructional Technique (Time in min)			Method	Refs	Comments
		T	D	P			
TP1							
CAT diesel generator set: 1. Location, function, general overview; 2. Modes of operation: a. Locations for start and stop; b. Emergency starts and stops; and c. Fail starts and stops. 3. Parameters: a. Alarms; b. Trips; and c. Warnings.		50	25		Lecture/ demo	A1, A4, A7, C6, C7, C8, C44, C45, C57, C58, C59	
TP2							
Electrical distribution system: 1. Location, function, general overview; 2. Switchboard: a. Programmable Logic Controller (PLC); and b. Configuration. 3. Power supplies and loads: a. 440v loads; b. 220v loads; c. 115v loads; d. 24v DC1 loads;		100	100		Lecture/ demo/ OJT	A1, A4, A7, C6, C7, C8, C39, C40, C44, C45, C57, C58, C59	

e. 24v DC2 loads; f. 12v Loads; and g. Outback Inverter/Rectifier/Chargers. 4. Modes of operation: a. Automatic; b. Manual; and c. Abnormal configuration. 5. Parameters: a. Alarms; b. Trips; and c. Warnings.						

Enabling check - EC 001.03 Explain the Power Generation and Distribution System (PG&D).

PART I – ADMINISTRATIVE INSTRUCTIONS

Material and Facilities Instructions

1. Classroom.

Arrangements and Procedures

2. EC 001.03 will be a written homework assessment based on content from EO 001.03 and reviewed during class time. EO 001.03 will be tested under PC 001.03.

PART II – INSTRUCTIONS GIVEN TO TRAINEE

1. EC 001.03 will be a written homework assessment based on content from EO 001.03.

2. Read each question and all responses carefully, before answering the question.

3. Written homework will be reviewed and verified in a group discussion setting.

4. Do you have any questions?

PART III – ENABLING CHECK NOTES

1. Instructor to verify homework completed prior to discussion group.

2. Trainees to ensure accuracy of answers based on group discussion led by instructor.

3. Written homework to become self-study aid for PC 001.03 and PC 004.

PART IV – ENABLING CHECK LIST

N/A.

PERFORMANCE CHECK - PC 001.03

PART I – ADMINISTRATIVE INSTRUCTIONS

Material and Facilities Instructions

1. Classroom.

Arrangements and Procedures

2. PC 001.03 will be a written test based on content from EO 001.01 through EO 001.03.

3. Pass mark will be 60%.

4. Time required: 2 x 50 min periods.

5. References will not be required.

PART II – INSTRUCTIONS GIVEN TO TRAINEE

1. PC 001.03 will be a written test based on content from EO 001.01 through EO 001.03.

2. Ensure your desk is clear of material except for the test booklet and/or equipment necessary for this exam.

3. Do not open the booklet until you are told to do so.

4. Write your name on the answer sheet and all scrap paper. Ensure all scrap paper is attached to the appropriate answer sheet.

5. Work carefully. Do not spend too much time trying to answer question(s) with which you have difficulty. Continue the exam then come back to the problem question(s).

6. Read each question carefully, and all responses fully, before answering the question.

7. If you have any questions during the exam, raise your hand.

8. References will not be provided.

9. Pass mark will be 60%.

10. You will have 2 x 50 min period to complete this exam.

11. Do you have any questions before you start?

PART III – PERFORMANCE CHECK NOTES

Examination provided by NTDC(P) MSE Division

PART IV – EDUCATIONAL CHECK LIST

N/A.

EO 001.04 Explain Shaft line.

1. Performance. Explain shaft line.

2. Conditions.

    a. Given:

        (1) Operational IPCS/Orca Class Vessel;

        (2) Schematics and illustrations from naval and manufacturers' manuals; and

        (3) References.

    b. Denied: Nil.

    c. Environmental: Orca Class Vessel Training Room.

3. Standard. The trainee shall, IAW references, explain the shaft line arrangement including:

    a. Shaft line system components; and

    b. Modes of operation.

4. Time.

    a. 2 x 50 min periods; and

    b. 1 x 25 min EC – homework assignment to be corrected in class.

5. Method / Media.

    a. Lecture – 2 x 50 min periods;

    b. Demonstration – time captured during lecture periods and during PO 003; and

    c. Ship tour.

6. References. A1, A2, A4, A5, C4, C5, C8, C9, C38, C57, C58, C59

7. Training Aids.

    a. Power Point presentations;

    b. System drawings;

    c. Orca Class Manual; and

    d. Orca Class Vessel.

8. Test Details. EC 001.04 will be a homework assignment to be corrected during class time. EO 001.04 will be tested in PC 001.07.

9. Remarks. Nil

10. Teaching Points.

Teaching Point	Sub Teaching Points	Instructional Technique (Time in min)			Method	Refs	Comments
		T	D	P			
TP1							
Shaft line system components: 1. Location, function, and general overview: a. Flexible coupling; b. Companion flange; c. Passive shaft grounding; d. Stern tube seal; e. Stern tube bearing; f. Propeller g. A Bracket bearing; and h. Propeller shaft.		25	25		Lecture/ demo	A1, A2, A4, A5, C4, C5, C8, C9, C38, C57, C58, C59	
TP2							
Modes of operation: 1. Normal; and 2. Abnormal;		25	25		Lecture/ demo	A1, A2, A4, A5, C4, C5, C8, C9, C38, C57, C58, C59	

Enabling check - EC 001.04 - Explain Shaftline Arrangements.

PART I – ADMINISTRATIVE INSTRUCTIONS

Material and Facilities Instructions

1. Classroom.

Arrangements and Procedures

2. EC 001.04 will be a written homework assessment based on content from EO 001.04 and reviewed during class time. EO001.04 will be tested under PC001.07.

PART II – INSTRUCTIONS GIVEN TO TRAINEE

1. EC 001.04 will be a written homework assessment based on content from EO 001.04.
2. Read each question carefully, and all responses fully, before answering the question.
3. Written homework will be reviewed and verified in a group discussion setting.
4. Do you have any questions before you start?

PART III – ENABLING CHECK NOTES

1. Instructor to verify homework completed prior to discussion group.
2. Trainees to ensure accuracy of answers based on group discussion led by instructor.
3. Written homework to become self-study aid for PC.

PART III – ENABLING CHECK NOTES

Give guidance for the development of the assessment.

PART IV – ENABLING CHECK LIST

N/A.

EO 001.05 Explain ZF Gearbox.

1. Performance. Explain main gearing.

2. Conditions.

    a. Given:

        (1) Operational IPCS/Orca Class Vessel;

        (2) Schematics and illustrations from naval and manufacturers' manuals; and

        (3) References.

    b. Denied: Nil.

    c. Environmental: ORCA Class Vessel Training Room.

3. Standard. The trainee shall, IAW references, explain main gearing including:

    a. Physical description and components; and

    b. Modes of operation.

4. Time.

    a. 2.5 x 50 min periods; and

    b. 1x 25 min EC – homework assignment to be corrected in class.

5. Method / Media.

    a. Lecture;

    b. Demonstration – time captured during lecture periods and during PO 003; and

    c. Ship tour.

6. References. A1, A2, A4, C4, C5, C8, C9, C44, C45, C57, C59, C59.

7. Training Aids.

    a. Power Point presentations;

    b. System drawings;

    c. Orca Class Manual; and

    d. Orca Class Vessel.

8. Test Details. EC 001.05 will be a homework assignment to be corrected during class time. EO 001.05 will be tested in PC 001.07

9. Remarks. Nil.

10. Teaching Points.

Teaching Point	Sub Teaching Points	Instructional Technique (Time in min)			Method	Refs	Comments
		T	D	P			
TP1							
Physical description and components – location, function and general overview: 1. ZF Gearbox layout; 2. ZF Gearbox components: a. Oil cooler; b. Gear driven oil pump; c. Autotroll controller; d. Reduction ratios; e. Electrical Transmission Actuator; f. Manual Select Lever for manual control; g. Oil drain; h. Input and output shaft connections; i. Oil filter; j. Trailing lubricating oil pump; k. Oil Chamber drain; l. Oil fill fitting. 3. Sea water cooling; 4. Shaft lock; and	50	25		Lecture/ demo	A1, A2, A4, C4, C5, C8, C9, C44, C45, C57, C59, C59	
TP2							
Modes of operation: 1. Normal; 2. Manual; and 3. Emergency	25	25		Lecture/ demo	A1, A2, A4, C4, C5, C8, C9, C44, C45, C57, C59, C59	

Enabling check - EC 001.05 - Explain ZF Gearbox.

PART I – ADMINISTRATIVE INSTRUCTIONS

Material and Facilities Instructions

1. Classroom.

Arrangements and Procedures

2. EC 001.05 will be a written homework assessment based on content from EO 001.05 and reviewed during class time. EO 001.05 will be tested under PC 001.07.

PART II – INSTRUCTIONS GIVEN TO TRAINEE

1. EC 001.05 will be a written homework assessment based on content from EO 001.05.

2. Read each question carefully, and all responses fully, before answering the question.

3. Written homework will be reviewed and verified in a group discussion setting.

4. Do you have any questions before you start?

PART III – ENABLING CHECK NOTES

1. Instructor to verify homework completed prior to discussion group.

2. Trainees to ensure accuracy of answers based on group discussion led by instructor.

3. Written homework to become self-study aid for PC.

PART III – ENABLING CHECK NOTES

Give guidance for the development of the assessment.

PART IV – ENABLING CHECK LIST

N/A.

EO 001.06 Explain the Main Engines (MEs).

1. Performance. Explain the MEs.

2. Conditions.

   a. Given:

      (1) Operational IPCS/Orca Class Vessel;

      (2) Schematics and illustrations from naval and manufacturers' manuals; and

      (3) References.

   b. Denied: Nil.

   c. Environmental: Orca Class Vessel Training Room

3. Standard. The trainee shall, IAW references, explain the PDE including:

   a. Physical description and components; and

   b. Modes of operation.

4. Time.

   a. 5.5 x 50 min periods; and

   b. 1 x 50 min EC – homework assignment to be corrected in class.

5. Method / Media.

   a. Lecture – 5.5 x 50 min periods;

   b. Demonstration – time captured during lecture periods; and

6. References. A1, A2, A4, C4, C5, C8, C9, C44, C45, C57, C58, C59.

7. Training Aids.

   a. Power Point presentations;

   b. System drawings;

   c. Orca Class Manual; and

   d. Orca Class Vessel.

8. Test Details. EC 001.06 will be a homework assignment to be corrected during class time. EO 001.06 will be tested in PC 001.07.

9. Remarks. Nil.

10. Teaching Points.

Teaching Point	Sub Teaching Points	Instructional Technique (Time in min)			Method	Refs	Comments
		T	D	P			
TP1 Physical description and components to include:							
1. Location, function and general overview; 2. Marine Power Display (MPD); 3. Fuel system; a. Fuel Filters b. Priming c. Fuel pump 4. Lube oil system; a. Filters b. Cooler c. Pump 5. Charge Air; a. Turbochargers b. Aftercoolers 6. 24(v) systems: a. Start batteries; b. Alternator; and c. ECM. 7. Intake and exhaust: a. Dampers; b. Demisters; c. Filters; and d. Drains. 8. Water cooling systems: a. Seawater pump; b. Plate cooler; c. Freshwater pump; d. Oil cooler; and e. Aftercooler. 9. Governors:		150	50			A1, A2, A4, C4, C5, C8, C9, C44, C45, C57, C58, C59.	
TP2 Modes of operation to include:							
Normal; Emergency; and Parameters: a. Alarms; b. Trips; and		50	25		IL DP	A1, A2, A4, C4, C5, C8, C9, C44, C45,	

c. Warnings.				C57, C58, C59.	

Enabling check - EC 001.06 Explain the Main Engine (ME).

PART I – ADMINISTRATIVE INSTRUCTIONS

Material and Facilities Instructions

1. Classroom.

Arrangements and Procedures

2. EC 001.06 will be a written homework assessment based on content from EO 001.06 and reviewed during class time. EO 001.06 will be tested under PC 001.07.

PART II – INSTRUCTIONS GIVEN TO TRAINEE

1. EC 001.06 will be a written homework assessment based on content from EO 001.06.

2. Read each question carefully, and all responses fully, before answering the question.

3. Written homework will be reviewed and verified in a group discussion setting.

4. Do you have any questions before you start?

PART III – ENABLING CHECK NOTES

1. Instructor to verify homework completed prior to discussion group.

2. Trainees to ensure accuracy of answers based on group discussion led by instructor.

3. Written homework to become self-study aid for PC.

PART III – ENABLING CHECK NOTES

Give guidance for the development of the assessment.

PART IV – ENABLING CHECK LIST

N/A

PERFORMANCE CHECK – PC 001.06

PART I – ADMINISTRATIVE INSTRUCTIONS

Material and Facilities Instructions
1. Classroom.

Arrangements and Procedures
1. PC 001.06 will be a written test based on content from EO 001.04 through EO 001.06.
2. Pass mark will be 70%.
3. Time required: 2 x 50 min periods.
4. References will not be required.

PART II – INSTRUCTIONS GIVEN TO TRAINEE

1. PC 001.06 will be a written test based on content from EO 001.04 through EO 001.06
2. Ensure your desk is clear of material except for the test booklet and/or equipment necessary for this exam.
3. Do not open the booklet until you are told to do so.
4. Write your name on the answer sheet and all scrap paper. Ensure all scrap paper is attached to the appropriate answer sheet.
5. Work carefully. Do not spend too much time trying to answer question(s) with which you have difficulty. Continue the exam then come back to the problem question(s).
6. Read each question carefully, and all responses fully, before answering the question.
7. If you have any questions during the exam, raise your hand.
8. References will not be provided.
9. Pass mark will be 70%.
10. You will have 2 x 50 min periods to complete this exam.
11. Do you have any questions before you start?

PART III – PERFORMANCE CHECK NOTES

Examination provided by NTDC(P) MS Div.

PO 002 OPERATE INTEGRATED PLATFORM CONTROL SYSTEM (IPCS) CONSOLE

EO 002.01 Operate Bridge Control Console

1. Performance. Operate Bridge control console.

2. Conditions.

   a. Given:

      (1) Operational IPCS/Orca Class Vessel;

      (2) Schematics and illustrations from naval and manufacturers' manuals; and

      (3) References.

   b. Denied: Nil.

   c. Environmental: Orca Class Vessel Training Room

3. Standard. The Engineering Watchkeeper shall independently operate the Bridge control console, IAW refs by:

   a. Supervising engineering pre-sail checks (CT0350);

   b. Operating IPCS man machinery interface (HT0040) to include:

      (1) Operating IPCS applications (HT0060);

      (2) Operating IPCS EHM (HT0070);

      (3) Operating IPCS software utilities (HT0050);

      (4) Testing components with built-in test equipment (BITE) (ALT1520)

      (5) Investigating system operation logs (AKT1090); and

      (6) Interpreting drawings/schematics (AMT0160).

   c. Managing damage control conditions (IT0380) to include Monitoring of damage control systems (IT0510).

4. Time Required.

   a. EO – 4.5 x 50 minute periods.

5. Instructional Methods.

a. Interactive lecture; and

b. Demonstration – practical.

6. Training Aids.

a. Power Point presentations;

b. System drawings;

c. Orca Class Manual; and

d. Orca Class Vessel.

7. Test Details. Trainees will be provided feedback throughout the EO. No formal EC or PC is included in this EO, instead the material will be tested in PC 002.

8. References. A1, A2, A4, A5, A7, C2, C3, C4, C5, C6, C7, C8, C9, C11, C12, C13, C14, C17, C18, C19, C20, C24, C31, C37, C38, C42, C44, C45, C57, C58, C59.

9. Remarks. Nil.

10. Teaching Points:

Teaching Point	Sub Teaching Points	Instructional Technique (Time in min)			Method	Refs	Comments
		T	D	P			
TP1 Supervise engineering pre-sail checks to include:							
a.	Supervising engineering pre-sail checks by: (1) Explaining Notice for Power; 1. Duties of the Engineering Watchkeeper; 2. Pre-watch rounds; and 3. Procedures for flashing up equipment.	50		25	IL	A1, A2, A4, A5, A7, C2, C3, C4, C5, C6, C7, C8, C9, C11, C12, C13, C14, C17, C18, C19, C20, C24, C31, C37, C38,	

					C42, C44, C45, C57, C58, C59.	
TP2 Operate IPCS main machinery interface to include:						
a. Operating IPCS applications by: (1) Navigating through pages; and (2) Utilizing maintenance pages	25	25		IL DP	A1, A2, A4, A5, A7, C2, C3, C4, C5, C6, C7, C8, C9, C11, C12, C13, C14, C17, C18, C19, C20, C24, C31, C37, C38, C42, C44, C45, C57, C58, C59.	
b. Operating IPMS EHM by: (1) Analyze trending.						
c. Operating IPCS software utilities						
d. Testing components with built-in test equipment						
e. Investigate system operation logs by: (1) Reviewing data captured in IPCS logs.						
f. Interpreting drawings/schematics by: (1) Understanding the IPCS interface; and (2) Relating the IPCS schematics to the ships official drawings.	25	25		IL DP		
TP3 Manage damage control conditions to include:						
a. Managing damage control conditions by: (1) Understanding appropriate damage control conditions; (2) Maintaining control over the watertight integrity board; (3) Managing damage control roundspersons during ship evolutions, i.e. Specials; and (4) Informing command of the current status of the ship's watertight integrity.	50			IL DP	A1, A2, A4, A5, A7, C2, C3, C4, C5, C6, C7, C8, C9, C11, C12, C13, C14, C17, C18,	

b. Monitoring of damage control system by: (1) Understanding how to activate the following damage control systems: i. Firemain; ii. CO2 Fire Suppression system; iii. Quick Closing Hydraulic Valves; iv. Ventilation Dampers; v. Ammo/pyro locker flooding; and vi. Crash stops (2) Understanding damage control alarms, to include: i. Verbal alarms; and ii. Automatic alarms.					C19, C20, C24, C31, C37, C38, C42, C44, C45, C57, C58, C59.	

EO 002.02 Operate Auxiliary Systems

1. Performance. Operate auxiliary systems.

2. Conditions.

    a. Given:

        (1) Operational IPCS/Orca Class Vessel;

        (2) Schematics and illustrations from naval and manufacturers' manuals; and

        (3) References.

    b. Denied: Nil.

    c. Environmental: Orca Class Vessel Training Room.

3. Standard. The Engineering Watchkeeper shall independently operate auxiliary systems, IAW refs by:

    a. Leading collective operation of auxiliary systems (ET0230) to include:

        (1) Operating steering system remotely (XT0090);

        (2) Operating fuel oil filling/transfer system remotely (DT0120);

    b. Monitoring ship's stability (WT0090) by:

        (1) Assessing ship's stability (WT0040);

        (2) Monitoring liquid level management system (CT0270);

        (3) Adjusting ship's stability (WT0010); and

        (4) Recording ship's stability variables (WT0110).

4. Time Required.

    a. EO – 6.5 x 50 minute periods.

5. Instructional Methods.

    a. Interactive lecture; and

    b. Demonstration – practical.

6. Training Aids.

    a. Power Point presentations;

b. System drawings;

c. Orca Class Manual; and

d. Orca Class Vessel.

7. Test Details. Trainees will be provided feedback throughout the EO. No formal EC is included in this EO, instead the material will be tested in PC 002.

8. References. A1, A2, A4, A5, A6, A7, C11, C44, C45, C57, C58, C59.

9. Remarks. Nil.

10. Teaching Points:

Teaching Point	Sub Teaching Points	Instructional Technique (Time in min)			Method	Refs	Comments
		T	D	P			
TP1 Lead collective operation of auxiliary systems to include:							
a. Operating steering system remotely.				50	DP	A1, A2, A4, A5, A6, A7, C11, C44, C45, C57, C58, C59	
b. Operating fuel oil filling/transfer system to include: (1) Transferring fuel by fuel transfer pump; and (2) Operating the transfer pump remotely. (3) Transfer fuel using hand pump				50			
TP2 Monitor ship's stability to include:							
a. Monitoring liquid level management system by: (1) Monitoring the ship's fluid systems, to include: i. Fresh water; ii. Black and grey water; iii. Diesel fuel; iv. Lube oil. v. Dirty bilge water vi. Clean bilge water vii. Dirty oil			25	50	IL DP	A1, A2, A4, A5, A6, A7, C11, C44, C45, C57, C58, C59	
b. Assessing ship's stability by: (1) Explaining stability terminology;			100				

(2) Interpreting drafts; (3) Monitoring liquid levels in damaged compartments due to flooding.				
c. Adjusting ship's stability by: (1) Transferring liquids between tanks; (2) Embarking liquids, to include: i. Fuels; ii. Fresh water.	25	25	A1, A2, A4, A5, A6, A7, C11, C44, C45, C57, C58, C59	

EO 002.03 Lead Collective Operation of Ancillary Systems (DT0060)

1. Performance. Lead collective operation of ancillary systems.

2. Conditions.

    a. Given:

        (1) Operational IPCS/Orca Class Vessel;

        (2) Schematics and illustrations from naval and manufacturers' manuals; and

        (3) References.

    b. Denied:

    c. Environmental: Orca Class Vessel Training Room.

3. Standard. The Engineering Watchkeeper shall independently lead collective operation of ancillary systems (DT0060) on the IPMS console, IAW refs, by:

    a. Operating ancillary systems (DT0070) to include:

        (1) Operating seawater cooling system. (DT0200);

        (2) Operating Fuel Oil Service (FOS). (DT0120)

4. Time Required.

    a. EO – 2 x 50 minute periods.

5. Instructional Methods.

    a. Interactive lecture; and

    b. Demonstration – practical.

6. Training Aids.

    a. Power Point presentations;

    b. System drawings;

    c. Orca Class Manual; and

    d. Orca Class Vessel.

7. Test Details. Trainees will be provided feedback throughout the EO. No formal EC is included in this EO, instead the material will be tested in PC 002.

8. References. A1, C2, C3, C44, C45, C57, C58.

9. Remarks. Nil.

10. Teaching Points:

Teaching Point	Sub Teaching Points	Instructional Technique (Time in min)			Method	Refs	Comments
		T	D	P			
TP1 Operating ancillary systems to include:							
a. Operating Sea Water cooling system.				50	DP	A1, C2, C3, C44, C45, C57, C58	
b. Operating FOS system to include: (1) 3 Way Ganged Valves; and (2) Managing tank levels.			25	25	DP		

EO 002.04 Operate Domestic Systems (DT0060)

1. Performance. Lead collective operation of ancillary systems.

2. Conditions.

    a. Given:

        (1) Operational IPCS/Orca Class Vessel;

        (2) Schematics and illustrations from naval and manufacturers' manuals; and

        (3) References.

    b. Denied:

    c. Environmental: Orca Class Vessel Training Room.

3. Standard. The Engineering Watchkeeper shall independently lead collective operation of ancillary systems (DT0060) on the IPMS console, IAW refs, by:

    a. Operating Domestic systems to include:

        (1) Operating Fresh Water System;

        (2) Operating Black water and Sanitary System;

        (3) Operating Grey Water System

        (4) Operating HVAC System

4. Time Required.

    a. EO – 3 x 50 minute periods.

5. Instructional Methods.

    a. Interactive lecture; and

    b. Demonstration – practical.

6. Training Aids.

    a. Power Point presentations;

    b. System drawings;

    c. Orca Class Manual; and

    d. Orca Class Vessel.

7. Test Details. Trainees will be provided feedback throughout the EO. No formal EC is included in this EO, instead the material will be tested in PC.

8. References. A1, C2, C3, C44, C45, C57, C58.

9. Remarks. Nil.

10. Teaching Points:

Teaching Point	Sub Teaching Points	Instructional Technique (Time in min)			Method	Refs	Comments
		T	D	P			
TP1 Operating Domestic systems to include:							
a. Operating Fresh Water System System				25			
b. Operating Black water and Sanitary System				50			
c. Operating Grey Water System				25			
d. Operating HVAC System				50			

EO 002.05 Operate Power Generation & Distribution (PG&D) System

1. Performance. Operate PG&D system.

2. Conditions.

    a. Given:

        (1) Operational IPCS/Orca Class Vessel;

        (2) Schematics and illustrations from naval and manufacturers' manuals; and

        (3) References.

    b. Denied:

    c. Environmental: Orca Class Vessel Training Room.

3. Standard. The Engineering Watchkeeper shall independently operate the PG&D system, IAW refs by:

    a. Operating generators in all modes (CT0280); and

    b. Configuring ship's electrical load (CT0060).

4. Time Required.

    a. EO – 3 x 50 minute period.

5. Instructional Methods. Demonstration – practical.

6. Training Aids.

    a. Power Point presentations;

    b. System drawings;

    c. Orca Class Manual; and

    d. Orca Class Vessel.

7. Test Details. Trainees will be provided feedback throughout the EO. No formal EC is included in this EO, instead the material will be tested in PC 002.

8. References. A1, A2, A4, A7, C6, C7, C38, C39, C40, C44, C45, C57, C58, C59.

9. Remarks. Nil.

10. Teaching Points:

Teaching Point	Sub Teaching Points	Instructional Technique (Time in min)			Method	Refs	Comments
		T	D	P			
TP1 Operate generators in all modes to include:							
a. Operating generators in all modes by: (1) Selecting automatic mode; (2) Selecting manual mode;			25	50	DP	A1, A2, A4, A7, C6, C7, C38, C39, C40, C44, C45, C57, C58, C59	
TP2 Configure ship's electrical load to include:							
a. Configuring ship's electrical load by: (1) Switching the power supplies between two Diesel Generators (DG's) in: i. Automatic; ii. Manual;			25	50	DP	A1, A2, A4, A7, C6, C7, C38, C39, C40, C44, C45, C57, C58, C59	

EO 002.06 Operate ZF Gear Box & Shaftline Systems

1. Performance. Operate gearing & shafting systems.

2. Conditions.

    a. Given:

        (1) Operational IPCS/Orca Class Vessel;

        (2) Schematics and illustrations from naval and manufacturers' manuals; and

        (3) References.

    b. Denied:

    c. Environmental: Orca Class Vessel Training Room

3. Standard. The Engineering Watchkeeper shall independently operate the gearing and shafting system, IAW refs by:

    a. Leading operation of ZF gear box and shaftline systems (ABT0050) to include:

        (1) Operating ZF gearbox and shaftline systems (ABT0170);

        (2) Testing operation of gearbox (BT0200);

        (3) Operating gearbox remotely (ABT0150);

        (4) Monitoring shaft line operation (ABT0080); and

4. Time Required.

    a. EO – 2 x 50 minute periods.

5. Instructional Methods.

    a. Interactive lecture; and

    b. Demonstration – practical.

6. Training Aids.

    a. Power Point presentations;

    b. System drawings;

    c. Orca Class Manual; and

    d. Orca Class Vessel.

7. Test Details. Trainees will be provided feedback throughout the EO. No formal EC is included in this EO, instead the material will be tested in PC 002.

8. References. A1, A2, A4, C4, C5, C8, C9, C44, C45, C57, C58.

9. Remarks. Nil.

10. Teaching Points:

Teaching Point	Sub Teaching Points	Instructional Technique (Time in min)			Method	Refs	Comments
		T	D	P			
TP1 Lead operation of gearing and shafting systems to include:							
a. Operating main gearing and shafting systems by: (1) Monitor Towing Operations; (2) Monitoring single shaft operations; and (3) Steer by Main Engines.			25		DP	A1, A2, A4, C4, C5, C8, C9, C44, C45, C57, C58.	
b. Testing operation of gearbox by: (1) Testing functionality of the transmission shifting mechanism;			25		DP		
d. Monitoring shaft line operation.			25		DP		.
e. Monitoring gearbox operation remotely.			25		DP		

EO 002.07 Operate Main Propulsion System

1. Performance. Operate main propulsion system.

2. Conditions.

    a. Given:

        (1) Operational IPCS/Orca Class Vessel;

        (2) Schematics and illustrations from naval and manufacturers' manuals; and

        (3) References.

    b. Denied:

    c. Environmental: Orca Class Vessel Training Vessel.

3. Standard. The Marine Technician Watchkeeper shall operate the main propulsion system independently, IAW refs by:

    a. Operating main propulsion system in all modes (RT0010) by:

        (1) Operating propulsion control remotely (ABT0210) to include:

            i. Operating Main Engine and Gear box remotely (QT0390).

        (2) Operating propulsion control locally (ABT0200) to include:

            i. Operating Main Engine and Gear box locally (QT0350).

4. Time Required.

    a. EO – 6 x 50 minute periods.

5. Instructional Methods.

    a. Interactive lecture;

    b. Role playing; and

    c. Demonstration – practical.

6. Training Aids.

    a. Power Point presentations;

    b. System drawings;

    c. Orca Class Manual; and

d. Orca Class Vessel.

7. Test Details. Trainees will be provided feedback throughout the EO. No formal EC is included in this EO, instead the material will be tested in PC 002.

8. References. A1, A2, A4, C4, C5, C8, C9, C44, C45, C57, C58.

9. Remarks. Nil

10. Teaching Points:

Teaching Point	Sub Teaching Points	Instructional Technique (Time in min)			Method	Refs	Comments
		T	D	P			
TP1 Operate the main engines in all modes to include:							
a. Operating the main propulsion system remotely by: (1) Operating the main engine remotely to include; and i. Engine prestart requirements; ii. Automatic engine start procedures; iii. Single engine operation; iv. Two engine operation; and v. Automatic engine shut down procedures.		25	25	100	IL DP RP	A1, A2, A4, C4, C5, C8, C9, C44, C45, C57, C58.	
b. Operating the main propulsion system locally by: (1) Operating the main Diesel Engine locally to include: i. Engine prestart requirements; ii. Manual engine start procedures; and		25	25	100	IL DP RP		Instructor to demonstrate starting of engines from LOPs.

PERFORMANCE CHECK – PC 002

PART I – ADMINISTRATIVE INSTRUCTIONS

Material and Facilitation Instructions

1. Orca Class IPCS Console.

2. Invigilators.

Arrangements and Procedures

3. PC 002 will be a practical assessment based on content from PO 002.

4. Trainees required to attain 76 points.

5. Time required: 1 x 50 minute periods.

6. References will be available.

PART II – ASSESSMENT INSTRUCTIONS GIVEN TO TRAINEE

PRACTICAL ASSESSMENT

1. PC 002 will be a practical assessment based on content from PO 002.

2. During this practical assessment, you will be required to operate IPCS console. You must operate IPCS console in accordance with the standard.

3. References will be provided.

4. Trainees required to attain 76 points.

5. You will have 50 minutes to complete this assessment.

6. Do you have any questions before you start?

PART III – PERFORMANCE CHECK NOTES

Give guidance for the development of the assessment.

1. See instructions to the evaluator located in the table below.

PART IV – PERFORMANCE CHECK LIST

Check List PC 002	SAT	UN-SAT	Comments (for invigilator, before printing this check list these comments to be deleted and this space used for invigilator comments).
Flash-up the ship from 4hrs NFP using the trainer and bring to immediate notice for power by:			
Conducting pre-sail checks.			The trainee is to use the check off list provided to go through all systems and prepare the ship for flash-up.
Conducting **SAFETY** brief with the flash-up watch.			This is a critical check off, if the trainee does not provide the proper brief it is considered a failure of this EC. Brief to be clear to all involved personnel.
Navigating through pages.			Does the trainee know how to effectively use the IPCS to navigate through all the pages?
Changing station in control.			Have the trainee pass and offer station in control twice.
Flashing-up auxiliary equipment.			The trainee is to flash-up auxiliary equipment in support of flashing up the plant to immediate notice for power.
Flashing-up ancillary equipment.			The trainee is to flash-up ancillary equipment.
Flashing Up Domestic Systems			The trainee is to flash-up domestic equipment.
Preparing Main Engines and Gear Box for flash-up.			The trainee to prepare the main engines for flash-up.
Bringing the ME to idle.			Bring the ME to idle.

Instructions to Evaluator:

1. The instructor is to have the trainee flash-up the ship using the trainer and bring to immediate notice for power;
2. The trainee shall meet each requirement in the checklist. If the trainee fails to provide a proper safety brief to the flash-up watch, it will be considered a fail;
3. Trainees begin with 100 points; each fault subtracts 7 from the total; and
4. The trainee must receive more than 75 points in total to pass this PC.

The only grade awarded for this PC is PASS (for totals 76 points or greater) or FAIL (for totals less than 76 points).

PO 003 INVESTIGATE ENGINEERING SYSTEMS

EO 003.01 Investigate collective operation of auxiliary systems (CT0130)

1. Performance. Investigate collective operation of auxiliary systems.

2. Conditions.

   a. Given:

      (1) Operational IPCS/Orca Class Vessel;

      (2) Schematics and illustrations from naval and manufacturers' manuals; and

      (3) References.

   b. Denied:

   c. Environmental: Orca Class Vessel.

3. Standard. The Engineering Watchkeeper shall independently investigate engineering systems, IAW refs by:

   a. Troubleshooting auxiliary diesel engine internal components (AVT1330), to include:

      (1) Troubleshooting auxiliary diesel engine control system (AVT1290);

      (2) Troubleshooting auxiliary diesel engine sea water cooling system (AVT1360);

      (3) Troubleshooting auxiliary diesel engine exhaust system (AVT1300);

      (4) Troubleshooting auxiliary diesel engine starting system (AVT1350);

      (5) Troubleshooting auxiliary diesel engine fresh water system (AVT1320);

      (6) Troubleshooting auxiliary diesel engine fuel system (AVT1310);

      (7) Troubleshooting auxiliary diesel engine lube oil system (AVT1340); and

      (8) Troubleshooting auxiliary diesel engine combustion air system (AVT1280).

   b. Troubleshooting sea water cooling system (AHT0830);

   c. Troubleshooting firemain (AHT0890); and

   d. Troubleshooting bilge system (AHT0840).

4. Time Required. EO –9 x 50 minute periods.

5. Instructional Methods.

   a. Interactive lecture;

   b. Role playing; and

   c. Demonstration – practical.

6. Training Aids.

   a. Power Point presentations;

   b. System drawings;

   c. Orca Class Manual; and

   d. Orca Class Vessel.

7. Test Details. EO 003.01 will be assessed in PC 004

8. References. A1, A2, A4, C7, C9, C10, C11, C13, C14, C15, C16, C31, C37, C44, C45, C48, C57, C58, C9.

9. Remarks. Nil.

10. Teaching Points.

Teaching Point	Sub Teaching Points	Instructional Technique (Time in min)			Method	Refs	Comments
		T	D	P			
TP1 Troubleshoot auxiliary diesel engine internal components to include:							
a.	Rectifying diesel engine alarms by: (1) Troubleshooting auxiliary diesel engine control system; (2) Troubleshooting auxiliary diesel engine sea water cooling system; (3) Troubleshooting auxiliary diesel engine exhaust system; (4) Troubleshooting auxiliary diesel engine starting system; (5) Troubleshooting auxiliary diesel engine fresh water system; (6) Troubleshooting auxiliary diesel engine fuel system;	200		100	IL DP RP	A1, A2, A4, C7, C9, C10, C11, C13, C14, C15, C16, C31, C37, C44, C45, C48,	Instructor to assume role as command for student briefing in practical phase.

(7) Troubleshooting auxiliary diesel engine lube oil system; and (8) Troubleshooting auxiliary diesel engine combustion air system.				C57, C58, C9	
TP2 Troubleshoot seawater cooling system to include:					
a. Assessing reasons for possible system pressure loss to include: (1) Pipe rupture; and (2) Clogged strainer. b. Determining Effects on other machinery.	25	25	IL DP RP		Instructor to assume role as command for student briefing in practical phase.
TP3 Troubleshoot firemain system to include:					
e. Assessing reasons for possible system pressure loss to include: (1) Pump failure; (2) Clogged strainer; (3) Firemain rupture;	25	25	IL DP RP		Instructor to assume role as command for student briefing in practical phase.
TP4 Troubleshoot bilge system to include:					
a. Managing the oil pollution and abatement system to include consolidating bilges.	25	25	IL DP		

EO 003.02 Investigate collective operation of ancillary systems (CT0120)

1. Performance. Investigate collective operation of ancillary systems.

2. Conditions.

   a. Given:

      (1) Operational IPCS/Orca Class Vessel;

      (2) Schematics and illustrations from naval and manufacturers' manuals; and

      (3) References.

   b. Denied: Nil.

   c. Environmental: Orca Class Vessel.

3. Standard. The Engineering Watchkeeper shall independently investigate engineering systems, IAW refs by:

   a. Troubleshooting ancillary systems (AHT0820) to include:

      (1) Troubleshooting seawater cooling system (AHT0880);and

      (2) Troubleshooting CLO filling and transfer system (AHT0850).

4. Time Required. EO – 2 x 50 minute periods.

5. Instructional Methods.

   a. Interactive lecture;

   b. Role playing; and

   c. Demonstration – practical.

6. Training Aids.

   a. Power Point presentations;

   b. System drawings;

   c. Orca Class Manual; and

   d. Orca Class Vessel.

7. Test Details. EO 003.02 will be assessed in PC 004.

8. References. A1, A2, A4, C7, C9, C10, C11, C13, C14, C15, C16, C31, C37, C44, C45, C48, C57, C58, C9.

9. Remarks. Nil.

10. Teaching Points:

Teaching Point	Sub Teaching Points	Instructional Technique (Time in min)			Method	Refs	Comments
		T	D	P			
TP1 Troubleshoot ancillary systems to include:							
a. Troubleshooting seawater cooling system to include: (1) Monitoring equipment temperature; (2) Identifying low flow conditions; (3) Identifying pipe rupture; and (4) Identifying clogged strainer.		25	25		IL DP	A1, A2, A4, C7, C9, C10, C11, C13, C14, C15, C16, C31, C37, C44, C45, C48, C57, C58, C9	
b. Troubleshooting CLO system by: (1) Troubleshooting CLO fill and transfer to include: i. Managing tank levels; ii. Transferring CLO; and		25	25		IL DP RP		Instructor to assume role as command for student briefing in practical phase.

EO 003.03 Investigate collective operation of power generation and distribution system

1. Performance. Investigate collective operation of PG&D system.

2. Conditions.

   a. Given:

      (1) Operational IPCS/Orca Class Vessel;

      (2) Schematics and illustrations from naval and manufacturers' manuals; and

      (3) References.

   b. Denied: Nil.

   c. Environmental: Orca Class Vessel.

3. Standard. The Engineering Watchkeeper shall independently investigate engineering systems, IAW refs by:

   a. Investigating collective operation of power generation and distribution system (CT0150) to include:

      (1) Investigating switchboard auto functions; and

      (2) Configuring ship's electrical load.

4. Time Required. EO – 2 x 50 minute periods.

5. Instructional Methods.

   a. Interactive lecture; and

   b. Demonstration – practical.

6. Training Aids.

   a. Power Point presentations;

   b. System drawings;

   c. Orca Class Manual; and

   d. Orca Class Vessel.

7. Test Details. EO 003.03 will be assessed in PC 004.

8. References. A1, A2, A4, C7, C9, C10, C11, C13, C14, C15, C16, C31, C37, C44, C45, C48, C57, C58, C9.

9. Remarks. Nil.

10. Teaching Points.

Teaching Point	Sub Teaching Points	Instructional Technique (Time in min)			Method	Refs	Comments
		T	D	P			
TP1 Investigate engineering systems to include:							
a.	Investigating collective operation of power generation and distribution system (CT0150) by: (1) Investigating switchboard auto functions; (2) Investigating electrical pages to include: and i. PG&D; ii. UPS; (3) Describing effects on other systems (can include CSE equipment).	50	50		IL DP	A1, A2, A4, C7, C9, C10, C11, C13, C14, C15, C16, C31, C37, C44, C45, C48, C57, C58, C9	

EO 003.04 Investigate operation of gearbox and shaftline systems (CT0160)

1. Performance. Investigate operation of gearing and shafting systems.

2. Conditions.

   a. Given:

      (1) Operational IPCS/Orca Class Vessel;

      (2) Schematics and illustrations from naval and manufacturers' manuals; and

      (3) References.

   b. Denied:

   c. Environmental: Orca Class Vessel.

3. Standard. The Engineering Watchkeeper shall independently investigate operation of gearing and shafting systems, IAW refs by:

   a. Evaluating gearbox function (BT0050); and

   b. Evaluating shaftline function (BT0060).

4. Time Required. EO – 2 x 50 minute periods.

5. Instructional Methods.

   a. Interactive lecture; and

   b. Demonstration – practical.

6. Training Aids.

   a. Power Point presentations;

   b. System drawings;

   c. Orca Class Manual; and

   d. Orca Class Vessel.

7. Test Details. EO 003.04 will be assessed in PC 004.

8. References. A1, A2, A4, C7, C9, C10, C11, C13, C14, C15, C16, C31, C37, C44, C45, C48, C57, C58, C9.

9. Remarks. Nil.

10. Teaching Points.

Teaching Point	Sub Teaching Points	Instructional Technique (Time in min)			Method	Refs	Comments
		T	D	P			
TP1 Evaluate gearbox function to include:							
a. Troubleshooting high temperatures by: (1) Investigating gearbox cooling.			50		DP	A1, A2, A4, C7, C9, C10, C11, C13, C14, C15, C16, C31, C37, C44, C45C48, C57, C58, C9	.
TP2 Evaluate shaftline function to include:							
a. Troubleshooting stern tube seal temperatures by: (1) Investigating stern tube cooling. (2) Investigating Gland follower condition. (3) Investigate shaft line vibrations			50		DP		

EO 003.05 Investigate Main Propulsion System

1. Performance. Investigate main propulsion system.

2. Conditions.

   a. Given:

      (1) Operational IPCS/Orca Class Vessel;

      (2) Schematics and illustrations from naval and manufacturers' manuals; and

      (3) References.

   b. Denied:

   c. Environmental: Orca Class Vessel.

3. Standard. The Engineering Watchkeeper shall independently investigate main propulsion systems, IAW refs by:

   a. Investigating collective operation of main propulsion systems (CT0140) to include:

      (1) Troubleshooting main propulsion systems (CT0430);

      (2) Troubleshooting fuel oil system (AHT0860);

4. Time Required. EO – 6.5 x 50 minute periods.

5. Instructional Methods.

   a. Interactive lecture; and

   b. Demonstration – practical.

6. Training Aids.

   a. Power Point presentations;

   b. System drawings;

   c. Orca Class Manual; and

   d. Orca Class Vessel.

7. Test Details. EO 003.05 will be assessed in PC 004.

8. References. A1, A2, A4, C7, C9, C10, C11, C13, C14, C15, C16, C31, C37, C44, C45, C48, C57, C58, C9.

9. Remarks. Nil.

10. Teaching Points.

Teaching Point	Sub Teaching Points	Instructional Technique (Time in min)			Method	Refs	Comments
		T	D	P			
TP1 Investigate collective operation of main propulsion systems to include:							
a.	Troubleshooting main propulsion system by: (1) Explaining engineering problems to include: i. Low voltage; ii. Low Engine oil pressure; iii. High coolant temperature; iv. Low Coolant temperature; v. Air inlet restrictions; vi. Engine over speed condition; vii. High Exhaust temperature; viii. High oil differential pressure; ix. High fuel filter differential pressure; x. High crank case pressure; xi. Low transmission oil pressure; xii. High transmission oil temperature; and xiii. Battle override switch.	100	100		IL DP	A1, A2, A4, C7, C9, C10, C11, C13, C14, C15, C16, C31, C37, C44, C45, C48, C57, C58, C9	
b.	Troubleshooting fuel oil service system:	25	50		IL DP		

PERFORMANCE CHECK – PO 003

The concepts in PO 003 will be tested in PC 004.

PO 004 RESPOND TO ALARMS AND EMERGENCIES

EO 004.01 Respond to Auxiliary and Ancillary System Alarms

1. Performance. Respond to auxiliary and ancillary system alarms.

2. Conditions.

   a. Given:

      (1) Operational IPCS/Orca Class Vessel;

      (2) Schematics and illustrations from naval and manufacturers' manuals; and

      (3) References.

   b. Denied: Nil.

   c. Environmental: Classroom/Orca Class Vessel.

3. Standard. The Engineering Watchkeeper shall respond to auxiliary and ancillary system alarms independently, IAW refs by:

   a. Responding to minor alarms and faults;

   b. Responding to auxiliary and ancillary emergencies IAW with Orca EOTIs and EOPs; and

   c. Advising command on auxiliary and ancillary systems by:

      (1) Advising command on HVAC air distribution system capabilities and limitations (JT0040)

      (2) Advising command on hotel system capabilities and limitations (JT0030).

4. Time Required.

   a. EO – 10 x 50 minute periods; and

   b. EC – 50 minutes per trainee.

5. Instructional Methods.

   a. Interactive lecture;

   b. Demonstration – practical; and

   c. Role playing.

6. Training Aids.

   a. Power Point presentations;

   b. System drawings;

   c. Orca Class Manual; and

   d. Orca Class Vessel.

7. Test Details. EO 004.01 will be tested in EC 004.01.

8. References. A1, A2, A4, C7, C9, C10, C11, C13, C14, C15, C16, C31, C37, C44, C45, C48, C57, C58, C9.

9. Remarks. Nil.

10. Teaching Points.

Teaching Point	Sub Teaching Points	Instructional Technique (Time in min)			Method	Refs	Comments
		T	D	P			
TP1 Respond to minor alarms and faults to include:							
a. Responding to auxiliary, ancillary and Domestic alarms and faults for the following: (1) Firemain; (2) Fresh water system; (3) Black water system; (4) Seawater cooling system; (5) Refrigeration system; (6) Clean lube oil; (7) Fuel oil service system; (8) Fuel filter water separator; (9) Oil pollution abatement system; (10) Compressed air system;			50	100	DP	A1, A2, A4, C7, C9, C10, C11, C13, C14, C15, C16, C31, C37, C44, C45, C48, C57, C58, C9	

(11) HVAC system; and (12) Damage control firefighting systems.						

TP2 Respond to auxiliary and ancillary emergencies by:						
a. Responding to the following auxiliary and ancillary emergencies IAW manual of emergency procedures, MSE(at sea for ORCA class ships): (1) Seawater cooling system; (2) Loss of main fuel system; and (3) Loss of Compressed Air.	50	100	100	IL DP		Trainees to receive a physical copy of Manual of Emergency
TP3 Advise command on auxiliary and ancillary systems by:						
a. Advising command on HVAC air distribution system capabilities and limitations (JT0040), to include the following topics: (1) Current system configuration; (2) Current system deficiencies; and (3) Possible solutions or work arounds to repair current deficiencies.		25	25	DP RP		
b. Advising command on Domestic system capabilities and limitations (JT0030), to include the following topics: (1) Fresh water system; (2) Grey Water system; and (3) Black water system.		25	25			

Enabling Check – EC 004.01

PART I – ADMINISTRATIVE INSTRUCTIONS

Material and Facilitation Instructions

1. MCR Trainer.

Arrangements and Procedures

2. EC 004.01 will be a practical assessment based on content from EO 004.01.

3. The trainee must achieve 76 points.

4. Time required: 50 minutes per trainee.

5. References will be available.

PART II – ASSESSMENT INSTRUCTIONS GIVEN TO TRAINEE

PRACTICAL ASSESSMENT

1. EC 004.01 will be a practical assessment based on content from EO 003.01, 003.02 and 004.01.

2. During this practical assessment, you will be required to respond to auxiliary and ancillary alarms. You must pass this test to pass EO 004.01. To pass the test, you must complete response to auxiliary and ancillary alarms in accordance with the standard.

3. References will be available.

4. The trainee must achieve 76 points.

5. IAW EC checklist

6. You will have 50 minutes to complete this assessment.

7. Do you have any questions before you start?

PART III – ENABLING CHECK NOTES

See instructions to the evaluator located in the table below.

PART IV – ENABLING CHECK LIST

Check List	SAT	UNSAT	Comments
Review engineering log.			The trainee is to review the engineering log and update as necessary.
Perform page rounds.			The trainee is to navigate IPCS pages and report any deficiencies. Trainee to take appropriate SIC and role attribution.
Effective Communication.			The trainee is to communicate to the Bridge, personnel, and OIC clearly and concisely.
Rectify diesel engine alarms.			Ensure diesel fault is added.
Respond to auxiliary and ancillary alarms and faults.			The trainee is to demonstrate understanding of the scenario cause and effect.
Respond to auxiliary and ancillary emergencies (**SAFETY**).			
Advise command on HVAC air distribution system capabilities and limitations.			
Advise command on Domestic system capabilities and limitations.			

Instructions to Evaluator:

1. The instructor is to have the trainee in the MCR trainer and respond to various procedures, briefs, faults, and alarms, including one emergency drill and three minor alarms;
2. The fault can be anything taught on the EO. Students are to receive one diesel fault and two other faults;
3. Trainees begin with 100 points; each ‘UNSAT’ subtracts 7 from the total;
4. The trainee must receive more than 75 points in total to pass this EC; and
5. The only grades awarded for this EC are PASS (for totals 76 points or greater) or FAIL (for totals less than 76 points).

EO 004.02 Respond to PG&D alarms

1. Performance. Respond to PG&D alarms.

2. Conditions.

   a. Given:

      (1) Operational IPCS/Orca Class Vessel;

      (2) Schematics and illustrations from naval and manufacturers' manuals; and

      (3) References.

   b. Denied:

   c. Environmental: Classroom/Orca Class Vessel.

3. Standard. The trainee shall respond to PG&D alarms independently, IAW refs by:

   a. Advising command on PG&D systems (CT0030) to include:

      (1) Advising command on ship's power configuration (PT0010);

      (2) Advising command on ship's electrical load (CT0050); and

      (3) Reporting on PG&D system (CT0330).

4. Time Required.

   a. EO – 3.5 x 50 minute periods; and

   b. EC – 50 minutes per trainee.

5. Instructional Methods.

   a. Role playing; and

   b. Demonstration – practical.

6. Training Aids.

   a. Power Point presentations;

   b. System drawings;

   c. Orca Class Manual; and

   d. Orca Class Vessel.

7. Test Details. EO 004.02 will be tested in EC 004.02.

8. References. A1, A2, A4, C7, C9, C10, C11, C13, C14, C15, C16, C31, C37, C44, C45, C48, C57, C58, C9

9. Remarks. Nil.

10. Teaching Points.

Teaching Point	Sub Teaching Points	Instructional Technique (Time in min)			Method	Refs	Comments
		T	D	P			
TP1 Respond to PG&D alarms to include:							
a. Advising command on PG&D systems (CT0030) by: (1) Advising command on ship's power configuration (PT0010); (2) Advising command on ship's electrical load (CT0050); and (3) Reporting on PG&D system (CT0330). b. Responding to emergencies: (1) High DG temperature (2) DG breaker trip			25	50	DP RP	A1, A2, A4, C7, C9, C10, C11, C13, C14, C15, C16, C31, C37, C44, C45, C48, C57, C58, C9	Instructor to assume role as command for student briefing in practical phase.
TP2 Respond to total power failure as an Engineering Watchkeeper by:							
a. Verifying total blackout conditions; b. Verifying auto sequences to include: (1) Generators auto sequence; (2) Emergency Steering considerations; and (3) Diesel driven fire pumps (DDFP). c. Verifying load shed breakers are reset; and				100	DP RP		

d. Restoring plant configuration.						

Enabling Check – EC 004.02

PART I – ADMINISTRATIVE INSTRUCTIONS

Material and Facilitation Instructions

1. ORCA Class Vessel.

Arrangements and Procedures

2. EC 004.02 will be a practical assessment based on content from EO 003.03 and 004.02.

3. Pass/Fail.

4. Time required: 50 minutes.

5. References will be available.

PART II – ASSESSMENT INSTRUCTIONS GIVEN TO TRAINEE

PRACTICAL ASSESSMENT

1. EC 004.02 will be a practical assessment based on content from EO 003.03 and 004.02.

2. During this practical assessment, you will be required to respond to PG&D alarms. You must pass this test to pass EO 004.02. To pass the test, you must respond to PG&D alarms in accordance with the standard.

3. References will be provided.

4. Pass/Fail IAW EC checklist.

5. You will have 50 minutes to complete this assessment.

6. Do you have any questions before you start?

PART III – ENABLING CHECK NOTES

See instructions to the evaluator located in the table below.

PART IV – ENABLING CHECK LIST

Check List	SAT	UNSAT	Comments
Review engineering log.			The trainee is to review the engineering log and update as necessary.

Perform page rounds.			The trainee is to navigate IPCS pages and report any deficiencies.
Effective Communication.			The trainee is to communicate to the MCR, personnel, and OOW clearly and concisely.
Respond to PG&D alarms and faults.			The trainee is to demonstrate understanding of the scenario cause and effect.
Respond to blackout			
Advise command on PG&D system capabilities and limitations.			

Instructions to Evaluator:

1. The instructor is to have the trainee in the MCR trainer and respond to various procedures, briefs, faults, and alarms, including one blackout;
2. The fault can be anything taught on the EO;
3. Trainees begin with 100 points; each ‘UNSAT’ subtracts 7 from the total;
4. The trainee must receive more than 75 points in total to pass this EC; and
5. The only grades awarded for this EC are PASS (for totals 76 points or greater) or FAIL (for totals less than 76 points).

EO 004.03 Respond to Main Propulsion Alarms

1. Performance. Respond to main propulsion alarms.

2. Conditions.

   a. Given:

      (1) Operational IPCS/Orca Class Vessel;

      (2) Schematics and illustrations from naval and manufacturers' manuals; and

      (3) References.

   b. Denied: Nil.

   c. Environmental: Orca Class Vessel.

3. Standard. The Engineering Watchkeeper shall respond to main propulsion alarms independently, IAW refs by:

   a. Advising command on machinery/system status (GT0010) to include: and

      (1) Advising command on propulsion capabilities and limitations (CT0040);

      (2) Advising command on marine systems operational capabilities and limitations (CT0020); and

      (3) Advising command on marine systems engineering parameters (CT0010).

   b. Reporting on propulsion deficiencies (CT0340) to include:

      (1) Reporting on machinery limitations (CT0310); and

      (2) Reporting on marine systems engineering parameters (CT0320).

4. Time Required.

   a. EO – 6 x 50 minute periods; and

   b. EC – 50 minutes per trainee.

5. Instructional Methods.

   a. Interactive lecture;

   b. Role playing; and

   c. Demonstration – practical.

6. Training Aids.

   a. Power Point presentations;

   b. System drawings;

   c. Orca Class Manual; and

   d. Orca Class Vessel.

7. Test Details. EO 004.03 will be assessed in EC 004.04.

8. References. A1, A2, A4, C7, C9, C10, C11, C13, C14, C15, C16, C31, C37, C44, C45, C48, C57, C58, C9.

9. Remarks. Nil.

10. Teaching Points.

Teaching Point	Sub Teaching Points	Instructional Technique (Time in min)			Method	Refs	Comments
		T	D	P			
TP1 Advise command on machinery/system status to include:							
a.	Advising command on propulsion capabilities and limitations by: (1) Adhering to power restrictions; (2) Impact on overall plant; (3) Explaining optimum engine speeds; and (4) Advising command on marine systems engineering parameters.	50	25	100	IL DP RP	A1, A2, A4, C7, C9, C10, C11,C13 C14,C15 C16,C31 C37,C44 C45,C48 C57,C58 C9	
b.	Advising command on marine systems operational capabilities and limitations to include explaining impact of failures or emergencies to the operation of the propulsion plant.	25	25	50	IL DP RP		
TP2 Report on propulsion deficiencies to include:							
Reporting on machinery limitations by reporting on marine systems engineering parameters; given the altered state of the plant due to emergencies/problems, what it means to the current capability.			25		DP RP	A1, A2, A4, C7, C9, C10, C11,C13 C14,C15 C16,C31	

				C37,C44 C45,C48 C57,C58 C9	

EO 004.04 Respond to Shipboard Emergencies

1. Performance. Respond to shipboard emergencies

2. Conditions.

   a. Given:

      (1) Operational IPCS/Orca Class Vessel;

      (2) Schematics and illustrations from naval and manufacturers' manuals; and

      (3) References.

   b. Denied: Nil.

   c. Environmental: Orca Class Vessel.

3. Standard. The Engineering Watchkeeper shall respond to shipboard emergencies independently, IAW refs by:

   a. Managing propulsion plant operation in emergency stations (IT0470); and

   b. Advising command on ship's stability (WT0020).

4. Time Required.

   c. EO – 1.5 x 50 minute periods; and

   d. EC – 200 minutes per trainee.

5. Instructional Methods.

   a. Interactive lecture;

   b. Role playing; and

   c. Demonstration – practical.

6. Training Aids.

   a. Power Point presentations;

   b. System drawings;

   c. Orca Class Manual; and

   d. Orca Class Vessel.

7. Test Details. EO 004.04 will be tested in EC 004.04.

8. References. A1, A2, A4, C7, C9, C10, C11, C13, C14, C15, C16, C31, C37, C44, C45, C48, C57, C58, C9.

9. Remarks. Nil.

10. Teaching Points.

Teaching Point	Sub Teaching Points	Instructional Technique (Time in min)			Method	Refs	Comments
		T	D	P			
TP1 Manage propulsion plant operation in emergency stations to include:							
a. Responding to the following ship's emergencies: (1) Emergency stations; (2) Fuel/HAZMAT spill; and (3) Rescue stations.		50			IL DP	A1, A2, A4, C7, C9,C10C 11,C13, C14,C15 C16,C31 C37,C44 C45,C48 C57,C58 C9	
TP2 Advise command on ship's stability to include:							
a. Recommending the following stability conditions by: (1) Transferring liquid loads; (2) Changing the ships course; and (3) Commencing pumping operations.		25			IL DP RP	A1, A2, A4, C7, C9,C10 C11,C13 C14,C15 C16,C31 C37,C44 C45,C48 C57,C58 C9	Instructor to assume role as command for student briefing in practical phase.

Enabling Check – EC 004.04

PART I – ADMINISTRATIVE INSTRUCTIONS

Material and Facilitation Instructions

1. Orca Class Vessel at Sea.

Arrangements and Procedures

2. EC 004.04 will be a practical assessment based on content from EO 003.04, 003.05, 004.03 and 004.04.

3. The trainee must achieve 76 points.

4. Time required: 150 minutes per student.

5. References will be available.

PART II – ASSESSMENT INSTRUCTIONS GIVEN TO TRAINEE

PRACTICAL ASSESSMENT

1. EC 004.04 will be a practical assessment based on content from EO 004.03 and 004.04.

2. During this practical assessment, you will be required to respond to main propulsion alarms and ship emergencies. You must pass this test to pass EO 004.03 and 004.04. To pass the test, you must respond to main propulsion alarms and ship emergencies in accordance with the standard.

3. References will be provided.

4. The trainee must achieve 76 points. IAW EC checklist

5. You will have 150 minutes to complete this assessment.

6. Do you have any questions before you start?

PART III – ENABLING CHECK NOTES

See instructions to the evaluator located in the table below.

PART IV – ENABLING CHECK LIST

Check List	SAT	UNSAT	Comments
Review engineering log.			The trainee is to review the engineering log and update as necessary.
Perform page rounds.			The trainee is to navigate IPCS pages and report any deficiencies.
Effective communication.			The trainee is to communicate with personnel and OIC in a clear and concise manner, showing confidence.
Manage propulsion plant operation in emergency stations.			Manage propulsion plant: - Emergency stations; - Rescue stations;and - Fuel/HAZMAT spill.
Advise command on propulsion capabilities and limitations.			The trainee is to explain to the OIC speed capabilities/limitations.
Advise command on marine systems operational capabilities and limitations.			The trainee is to explain operational impact of the following situations that can be selected: - Low Engine Voltage; - Low Engine Oil pressure; - High coolant temperature; - Low Coolant temperature; - Inlet air restriction; - Overspeed; - High exhaust temperature; - High air intake differential pressure; - Oil filter diff. pressure; - Fuel filter diff. pressure; - High crankcase pressure; - High transmission temp; and - Low transmission pressure.
Report on propulsion deficiencies.			Situation – Impact – Recommendation – Limitations standard of reporting
Report on machinery limitations.			Situation – Impact – Recommendation – Limitations standard of reporting
Advise command on ship's stability.			The trainee is to assess the ships current list and bring up the tanks overview page.

Instructions to Evaluator:

1. The instructor is to have the trainee on board ORCA Class Vessel and respond to various procedures, briefs, faults, and alarms to include three drills;
2. The scenarios selected should challenge the trainees knowledge and understanding of the systems; as long as it was taught on the EO;
3. Trainees begin with 100 points; each fault subtracts 7 from the total;
4. The trainee must receive more than 75 points in total to pass this EC; and
5. The only grades awarded for this EC are PASS (for totals 76 points or greater) or FAIL (for totals less than 76 points).

PERFORMANCE CHECK – PC 004

PART I – ADMINISTRATIVE INSTRUCTIONS

Material and Facilitate Instructions

1. ORCA Class Vessel at Sea.

Arrangements and Procedures

2. PC 004 will be a practical on-board assessment based on content from PO 001, PO 002, PO 003 and PO 004.

3. Pass/Fail.

4. Time required: 10 Days at sea on an operational ORCA Class Vessel.

5. References will be provided.

PART II – ASSESSMENT INSTRUCTIONS GIVEN TO TRAINEE

PRACTICAL ASSESSMENT

1. PC 004 will be a practical assessment based on content from PO 001, PO 002, PO 003 and PO 004.

2. During this practical assessment, you will be required to operate IPCS console. You must perform watchkeeping duties in accordance with the standard.

3. References will be provided.

4. Pass/Fail.

5. You will have six months at sea to complete this assessment.

6. This assessment will be completed upon being posted to an operational ORCA Class Vessel.

PART III – PERFORMANCE CHECK NOTES

The following checklist is a copy of Appendix 1 of Annex E which includes additional notes regarding assessment criteria. This checklist is to be used by the invigilator on-board ship to help with the assessment of the trainee.

PO/EO	Performance Statement	Comments
002.01	The Trainee shall Conduct Watchkeeping operations to the satisfaction of the Sr. Engineer by:	
002.01.001	Conduct pre-watch and on-watch rounds.	Sr. Engineer to conduct a set of rounds with the Trainee and be satisfied with the Trainee's knowledge
002.01.002	Conduct flash-up and shut-down routine.	Trainee to conduct pre-sail flash-up of the plant as ORCA Watchkeeper as well as shut down the plant at the end of a sail.
002.01.003	Operate propulsion system Remotely.	Trainee to prove Operate propulsion system Remotely to the satisfaction of the Sr. Eng.
002.01.004	Operate propulsion system locally.	Trainee to prove Operate propulsion system Locally to the satisfaction of the Sr. Eng.
002.01.005	Transferring fuel.	The Trainee is to transfer fuel from storage tank to day tanks and then from day tanks to storage tank.
002.04	The Trainee shall operate the PG&D to the satisfaction of the Sr. Eng by:	
002.04.001	Changing over electrical generators from the ER at sea.	The Trainee is to flash up and change over electrical load from on DG to the other at sea.
002.05	The Trainee shall operate the gear box to the satisfaction of the Sr. Eng by:	
002.05.001	Operating the Gearbox Locally.	The Trainee to utilize local control of gear box
002.05.002	Locking Shaftline and gearbox.	The Trainee to lock shaftline at sea.
004	The Trainee shall correctly perform delaying drills as ORCA Watchkeeper as per ORCA CLASS EOPs and EOTIs. All drills listed in the manual require a signature. If a drill is added/removed from the manual it is to be automatically added/removed from this OJPR.	

	All the drills below will be assessed by the Sr. Engineer.		
Loss of both shaft lines			
Loss of one shaft line			
Loss of station in control			
Loss of salt water cooling			
Propulsion engine high freshwater temperature			
Propulsion engine low lube oil pressure			
Propulsion engine inlet air restriction			
Propulsion engine high exhaust temperature			
Propulsion engine overspeed condition			
Propulsion engine high crankcase pressure			
Gearbox low lube oil pressure			
Gearbox high lube oil temperature			

Power failure			
Steering gear breakdown			
RTU#1 failure			
RTU #2 failure			
Respond to Fire			
Respond to Flood			
Respond to Man Over Board			
Diesel Generator Alarms			
Final set of drills containing no less than three continuous drills conducted by and to the satisfaction of the Sr. Engineer.			

PO 005 Perform the duties of Fuelling I/C

EO 005.01 Complete Fuelling Orders

1. Performance. Complete Fuelling Orders.

2. Conditions.

   a. Given:

      (1) Classroom;

      (2) PP&S;

      (3) Computer with DWAN access;

      (4) References; and

      (5) Scenarios with fuel spec sheets.

   b. Denied: Nil.

   c. Environmental: Classroom.

3. Standard. The trainee shall independently complete fuelling orders, IAW refs by:

   a. Advising command on fuel quality (OT0010) to include:

      (1) Formulating fueling recommendations.

   b. Adhering to SEMS SOPs by:

      (1) Describing fuelling party duties; and

      (2) Selecting fuelling checklists based on the fuelling evolution.

   c. Completing fuelling documentation to include:

      (1) Extracting data from fuel reports;

      (2) Initiating fuelling orders prior to fuelling evolutions; and

      (3) Finalizing fuelling orders after fuelling evolutions.

4. Time Required.

   a. EO – 2 x 50 mins to deliver training; and

   b. EC – OJT.

5. Instructional Methods.

   a. Guided discussion.

6. Training Aides. Nil.

7. Test Details. EO 002.01 will be tested in PC 005 IAW table below and Annex K

8. References. A2.

9. Remarks. Trainees have experience with SEMS SOP E1 and participating in fuelling evolutions in various positions. The purpose of this EO is to activate trainee experience and direct it specifically to applying SEMS in the completion of the checklist, not completing the checklist itself.

10. Teaching Points.

Teaching Points	Ref	Method / Media / Environment / Training Activity Details	Time (mins)
LEARNING OBJECTIVE: Complete fuelling orders IAW SEMs and other technical bulletins			
1. Advise command on fuel quality (OT0010): a. Formulating fuelling recommendations: i. Recommendation options; and ii. Impacts to recommendations. 2. Adhering to SEMS SOPs: a. Describing the duties of the fueling party IAW SEMS to include: i. Positions; and ii. Responsibilities. b. Explaining the types of fueling evolutions: i. Alongside; ii. Transferring fuel; and iii. De-fuelling. c. Describing fueling checklists: i. Fueling party; ii. Pre-fuelling; iii. Manual dip verification checklist to include: (1) Pre-fuelling; and (2) Post-fuelling. iv. Valve operation; v. Post-fuelling; and vi. Fueling report. 3. Completing fuelling documentation: a. Extracting data from fuel reports: i. Before and after fuelling; ii. Calculating fuel requirements; and iii. Calculate consolidation requirements. b. Initiating fuelling documentation prior to fuelling evolutions: i. Assigning personnel to fuelling party; ii. Determining tank filling sequence; and iii. Determining time requirements.	A2	**Method / Media:** F2F/Guided Discussion (GD) and follow on fuelling assignment **Trg Environment:** ORCA Classroom **Activity Details:** **A.** The instructor will lead a guided discussion on fuel quality, SEMS SOP and associated fuelling documentation. Trainees collectively have the knowledge and experience on these TPts and the guided discussion will allow trainee's experience to form the basis of instruction. Instructor will amplify important safety or points not addressed in trainee discussion. Questions for discussion should focus on: • Fuel quality; • Fuel types; • References and how to use them, specifically SEMS SOP, POL Handbook and Tech bulletins; • Fuelling party positions trainees have held and important points of those positions; • Key fuelling positions like the fuelling I/C and what those duties include; and • Fuelling documentation. **B.** Trainees will be given 20 scenario based questions and apply policy from SEMS in their answers. Questions will consist of questions on fuel quality, fuelling evolutions, calculating fuel quantity, fuel tank sequencing and assigning personnel to fuelling party positions. Scenarios should include alongside, at sea and transfer operations. **Trg Resources:** Guided Discussion: • instructions; and	100 min

Teaching Points	Ref	Method / Media / Environment / Training Activity Details	Time (mins)
c. Finalizing fuelling documentation after fuelling evolutions: i. Confirming amount of fuel received; ii. Submitting paperwork to OIC.		• Prepared guided discussion notes and questions; Fuelling Orders Activity: • Scenarios; • Calculator; • Access to references (SEMS SOP E1, Tech Bulletin, POL Handbook, Fuel Decision Tree); • List of personnel for scheduling fuelling party;	

PO 005 Perform the duties of Fuelling I/C

EO 005.02 Lead Maritime Vessel Fuelling

1. Performance. Lead Maritime Vessel Fuelling (internal fuel / providing / consolidating) (OT0090).

2. Conditions.

   a. Given:

      (1) ORCA Class Vessel at Fueling Jetty;

      (2) Role players as Watchkeepers and fuelling party;

      (3) Communications equipment;

      (4) References; and

      (5) Fuelling scenarios.

   b. Denied: Assistance.

   c. Environmental: ORCA Class Vessel at Fueling Jetty.

3. Standard. The trainee shall independently lead maritime vessel fuelling, IAW refs by:

   a. Preparing a ship to move (embark / disembark / transfer) fuel to include (NT):

      (1) Consolidating fuel tanks;

      (2) Briefing the fuel party; and

      (3) Configuring fuel system to receive fuel.

   b. Moving (embark / disembark / transfer) fuel to include:

      (1) Initiating fuelling evolutions;

      (2) monitoring flow rates;

      (3) Filling tanks IAW tank filling sequence; and

      (4) Stopping fuelling evolutions when tanks do not exceed:

         (a) 95% capacity alongside; or

         (b) 90% at sea.

   c. Securing a ship from fuelling operations to include (NT):

      (1) Reconfiguring fuel systems;

      (2) Securing fuelling equipment; and

      (3) Securing personnel.

4. Time Required.

   a. EO – 1.5 x 50 mins to deliver training

   b. EC – OJT.

5. Instructional Methods.

   a. Small group training activity.

6. Training Aides.

   a. ORCA vessel; and

   b. Fueling jetty.

7. Test Details. EO 005.02 will be tested in PC 005 IAW table below and Annex K.

8. References. A2.

9. Remarks. Nil.

10. Teaching Points.

Teaching Points	Ref	Method / Media / Environment / Training Activity Details	Time (mins)
LEARNING OBJECTIVE: Lead maritime vessel fuelling			
1. Moving (embark / disembark / transfer) fuel to include: a. Initiating fuelling evolutions: i. Dump set up: (1) Drip trays; (2) Fuel spill kits; and (3) Tools. ii. Communications checks: (1) Tank sentries; (2) Valve sentries; and (3) Dump personnel. iii. Tank filling: (1) operating locally and remotely; (2) valve operation order; iv. Path of orders to start/stop pumping: (1) Inform command ready to embark fuel. b. Monitoring flow rates: i. Confirming fuel is embarking by monitoring change in levels; and ii. Calculate rate of flow using change in volume over time.	A2 SOP E1	**Method / Media:** small group training activity **Trg Environment:** ORCA Class Vessel at Fueling Jetty. **Activity Details:** In groups, trainees will prepare and present a brief to the class based on the following parts of a fuelling evolution: • Initiating fuelling evolutions; • Monitoring flow rates; and • Filling tanks; and • Stopping fuelling evolutions. Briefings must include the teaching points and references associated with fuelling. Trainees should include real examples where possible. **Trg Resources:** • DWAN computer access; • Projector; • References; • Flipchart and paper;	30 min develop 40 min present (10 min/gp) 20 min debrief (5 min/gp)

Teaching Points	Ref	Method / Media / Environment / Training Activity Details	Time (mins)
c. Filling tanks IAW tank filling sequence to include: i. Ship's list; ii. Tank fill and stop timings based on the fuel rate; iii. operating locally and remotely; iv. valve operation order; v. Opening and closing tank valves; vi. Tank levels in isolated tanks; and vii. Manual dips. d. Maximum tank capacity: i. Maximum tank capacity: (1) At sea; and (2) Alongside. ii. Fuelling warnings: (1) 10 min; (2) 5 min; (3) 3 min; (4) 1 min; (5) Standby; and (6) Stop pumping. iii. Stopping pumping to include: (1) Maximum capacities alongside and at sea; (2) Tanks filled during normal fuelling; (3) Emergencies; (4) Blow down/gravity drain; and			

PC 005: Lead Vessel Fuelling

ASSESSMENT ACTIVITY: The trainee will calculate the amount of fuel required, consolidation plan and determine filling sequence, which the instructor will verify. Trainee will brief fuelling party IAW SEMS. Trainee will brief instructor on amount of fuel received. Trainees must not exceed filling levels detailed in checklist.

Pass/fail criteria: Pass/Fail IAW PC 005 checklist

ASSESSMENT RESOURCES:

- Operational Orca Class vessel Access references (SEMS SOP E1, Tech Bulletin, POL Handbook, Fuel Decision Tree)
- Access references (SEMS SOP E1, Tech Bulletin, POL Handbook, Fuel Decision Tree)
- List of personnel for scheduling fuelling party
- Ships fitted equipment
- Tank level tables
- Clock/stopwatch
- Standard fuel brief template

IMPACT OF FAILURE: Re-test / TRB IAW Ch 2 and NPTORD

ORCA Engineering Watchkeeper– PC 005 Assessment Checklist			
Trainee Name:	**Assessor Name:**	**Date:**	**Session:**

INVIGILATOR NOTES:

1. PASS / FAIL CRITERIA: Trainees must achieve a SAT on all Critical checklist items (marked with asterisk *) and achieve 70% SATs on the remainder of the checklist items to pass this assessment.
2. IMPACT OF FAILURE: Remedial training, re-test.

ITEM	SAT	UNSAT	COMMENTS
1. Prepare a ship to move (embark / disembark / transfer) fuel			
Provide recommendation to consolidate fuel tanks			
Brief the fuel party IAW template			
2. Move (embark / disembark / transfer) fuel			
Confirms inspection of dump to include: • Drip trays • Fuel spill kits • Tools			
Perform communications checks with: • Tank sentries • Valve sentries • Dump personnel			
Direct opening and closing tank filling valves locally and remotely			
Inform command ready to embark fuel			
3. Monitoring Flow Rate			
Confirms fuel is embarking by monitoring change in levels on IPCS			
Calculate rate of flow using change in volume over time • Direct entries in event log			
4. Filling tanks IAW tank filling sequence to include:			
Calculate tank fill and stop timings using fuel rate			
* Direct opening and closing tank valves as tanks fill by opening one before closing the previous one			
Monitor tank levels in isolated tanks to ensure levels aren't rising			
Confirm tank levels through manual dips • Dispatch tank sentries to verify readings on IPCS			
Correct actions taken when an abnormal event occurs • Stop fuelling • Based on abnormal event, explains to assessor: o the event o why fuelling was stopped o how to correct event o next actions after event corrected			
Makes appropriate recommendations in order to resume fuelling			

ITEM	SAT	UNSAT	COMMENTS
5. Stopping Fuelling Evolutions			
Announce fuelling warnings at:			
• 10 min;			
• 5 min;			
• 3 min;			
• 1 min;			
• Standby;			
• *Stop pumping.			
Stop pumping to include:			
• * No tanks exceed 95% alongside or 90% at sea;			
• Direct closing of tank valves locally.			

ADDITIONAL COMMENTS	
STRENGTHS:	**AREAS FOR DEVELOPMENT:**

THE CANDIDATE HAS RECEIVED A DEBRIEF AND SEEN THE CONTENT OF THIS ASSESSMENT			
OVERALL ASSESSMENT (circle one)		**FAIL**	**PASS**
Assessor	Name:	Date:	Signature:
Candidate	Name:	Date:	Signature:

ANNEX A - MAIN REFERENCES

A1 = CF or other official Canadian government publication

A2 = Second and subsequent references in any category

B = Official Allied Force or Allied government publication

C = Commercial publication

REF CODE	PUBLICATION	QTY
A1	PCTU EOTI's	1
A2	Naval Engineering Manual	1
A3	MARCORDs	1
A4	PCTU SOP's	1
A5	PCTU Bridge pack	1
A6	CANFLTPAC EMS	1
A7	General Electrical Specs for CF Ships	1
C1	Corrintec Installation, Operation and Maintenance Manual 'Minitek' Impressed Current Hull Protection System 20 AMP System (210-010)	1
C2	C-Shield Antifouling System Installation and Operation Manual. (210-010)	1
C3	C-Shield Log Sheet for Antifouling and Corrosion Control System (210-010)	1
C4	Gear Box & Shafting (FINNING)	1
C5	3516B Marine Engine Parts Manual (FINNING)	1
C6	3054 Marine Generator Set Parts Manual (FINNING)	1
C7	3054 Gen Set & Misc Manuals (FINNING)	1
C8	Drawings for 3054 & 3516B (FINNING)	1
C9	3516B Engine Manuals (FINNING)	1

REF CODE	PUBLICATION	QTY
C10	Technical Manual for Type EJ Seal (340-010)	1
C11	Steering System Installation and Service Manual (360-010)	1
C12	Black Water and Grey Water System Documentation for ORCA Class Vessels (420-010)	1
C13	Naval Training Vessel; Freshwater Pressure Set Grundfos Model CRN 1-5; Salt Water Pressure Set Grundfos Model CRN 1S5; HW Recirculation Pump Grundfos Model UPS15-42SF; Fuel Transfer Pump Viking Model HJ 190; Lube Oil Filling Pump Viking Model GX2-115; Oily Water/Waste Oil Eagle Model EP-33; Specialty Equipment Ltd. (420-010)	1
C14	YAG Navigation Training Vessel Installation Operation and Maintenance Manual Pumping Equipment (420-010)	1
C15	Installation and Operation Manual Hayward Model 53 BTX Duplex Strainer, Sizes ¾” – 4” (420-010)	1
C16	Liquid Controls Fuel Meter Model M5-A1 Owners Information Book (440-010)	1
C17	Technical Manual D200S Solas Davit S/N 2094-2099 (460-010)	1
C18	Muir VRC1100 – Vertical Windlass Installation and Instruction Manual (460-010)	1
C19	Delta “T” Systems 600-T4 Digital Ventilation Control System (480-010)	1
C20	Delta “T” Systems C! Ventilation Control System Installation and Operating Instructions (480-010)	1
C21	DX Remote and Self-Contained A/C Installation and Operation (480-010)	1
C22	SMX II Control Systems (DX) Installation (480-010)	1
C23	SMX II Control Systems (DX) Owner/Operator (480-010)	1
C24	Stork Marine Heating, Ventilation, Air Conditioning and Refrigeration Systems (480-010)	1
C25	Door and Window Documentation for ORCA Class Vessels (540-200)	1

REF CODE	PUBLICATION	QTY
C26	Office and Training Room Media Equipment for ORCA Class Vessels (540-300)	1
C27	Galley Equipment for ORCA Class Vessels (550-010)	1
C28	Laundry Equipment for ORCA Class Vessels (550-020)	1
C29	Inspection and Operation Booklet Radar Transponder Rescuer SART (560-010)	1
C30	Satfind-406 Pro Owners Manual (560-010)	1
C31	75/500FGX/MAX Series Fuel Filter/Water Separator for Diesel Engines (600-010)	1
C32	Installation Operation & Maintenance Manual Oily Water Separator (600-010)	1
C33	Instruction Manual 15ppm Bilge Alarm Type OMD-2005 (620-010)	1
C34	CO2 Carbon Dioxide Fire Suppression Systems – Owner's Manual (620-300)	1
C35	Dura-Power Commercial Electrical Water Heaters Models Dre-52, 80 & 120 Limited Warranty, Installation, Operation, Maintenance, Checklist (630-010)	1
C36	Halsey Taylor Owners Manual (630-010)	1
C37	Quincy R-25 Compressor Manuals (660-010)	1
C38	Additional Electrical Documentation for the ORCA Class Vessels (700-000)	1
C39	Thompson Technology Generator Control System (700-010)	1
C40	Outback Power Systems FX Series Inverter/Charger Documentation (710-100)	1
C41	Installation, Operation and Maintenance – 200 Watt Xenon Searchlight System Remote Electric Control (730-010)	1
C42	Fire Detection System for ORCA Class Vessels (740-010)	1

REF CODE	PUBLICATION	QTY
C43	Bridge Computer Documentation for ORCA Class Vessels (740-010)	1
C44	Integrated Platform Control System (IPCS) for the Canadian Navy YAG 320 Training Vessels – Operation and Maintenance Manual (740-010)	1
C45	Système intègre de commande de plateforme (SICP) pour les navires-ecoles YAG 320 de la Marine canadienne – Guide d'exploitations et de maintenance (750-010)	1
C46	External Communication Equipment for ORCA Class Vessels (750-010)	1
C47	Internal Communications Equipment for ORCA Class Vessels (750-010)	1
C48	Airchime Model KMJ-3 & WT-25700-c Serial # 40959 & 42299 (760-010)	1
C49	Instruction Manual of Automatic Digital Direction Finder TD-L 1360 – Operation, Installation and Maintenance (760-010)	1
C50	Marine Master Clock 100102-SP (760-010)	1
C51	Raytheon Marine AIS, DGPS, Meteo (760-010)	1
C52	Raytheon Marine Compass, Autopilot (760-010)	1
C53	Raytheon Marine Radar NSC18 (760-010)	1
C54	Raytheon Marine System Description for YAG-320 "ORCA Class" (760-010)	1
C55	Skipper DL850 2 Axis Doppler Log Operation and Installation Manual (760-010)	1
C56	Skipper GDS101 Graphic Depth Sounder Operation and Installation Manual (760-010)	1
C57	ORCA Class General Information Book – ENGLISH	1 per trainee
C58	ORCA Class General Information Book - FRENCH	1 per trainee

REF CODE	PUBLICATION	QTY
C59	Victoria Shipyards PCT Raven (Hull 56) Captain's Ship Book, Feb 2007, V.S.L. Ref NO. 05012	1
C60	Video: Fishing Boat Stability, VideoTel, Marine International. National Defence Canada. #02562. 1987	1

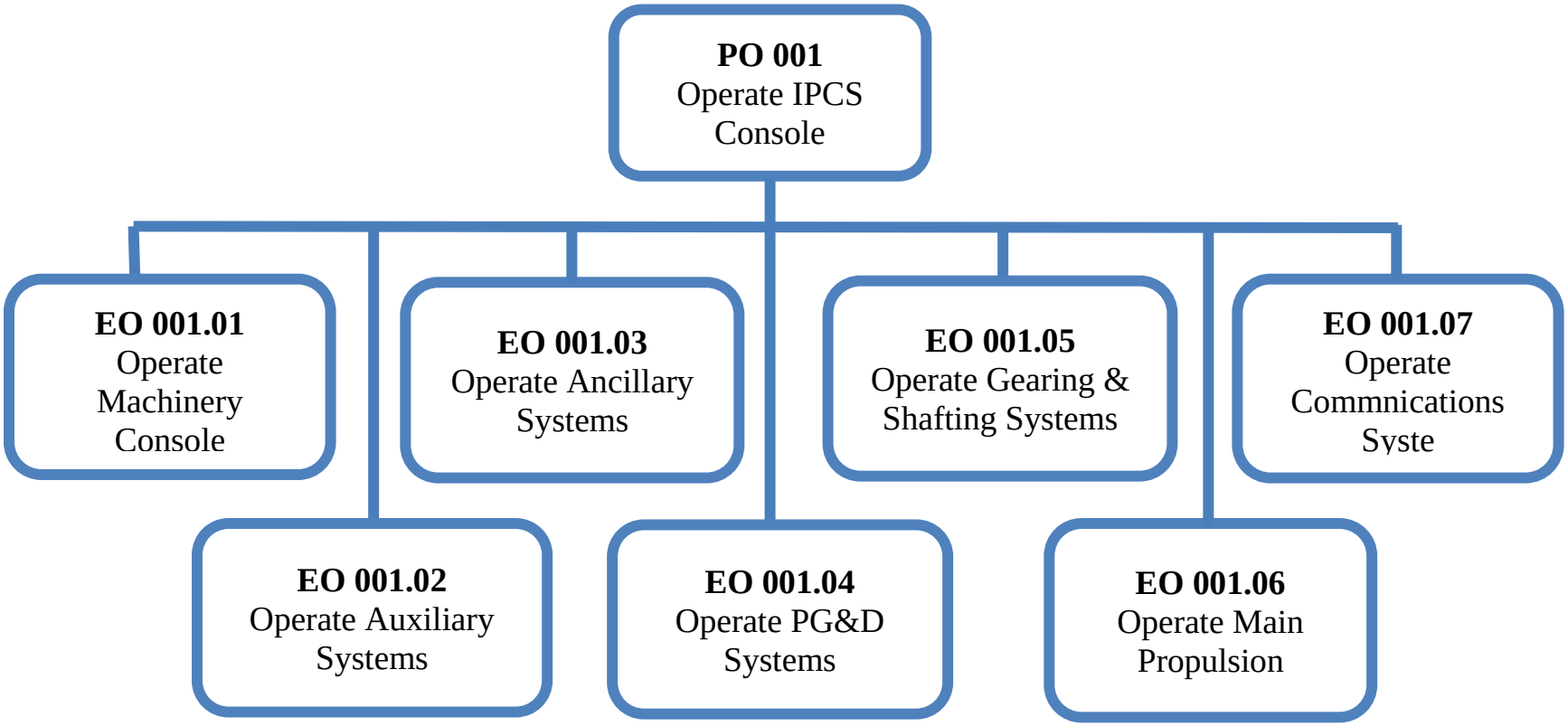
ANNEX B – TASK LIST (TRAIN)

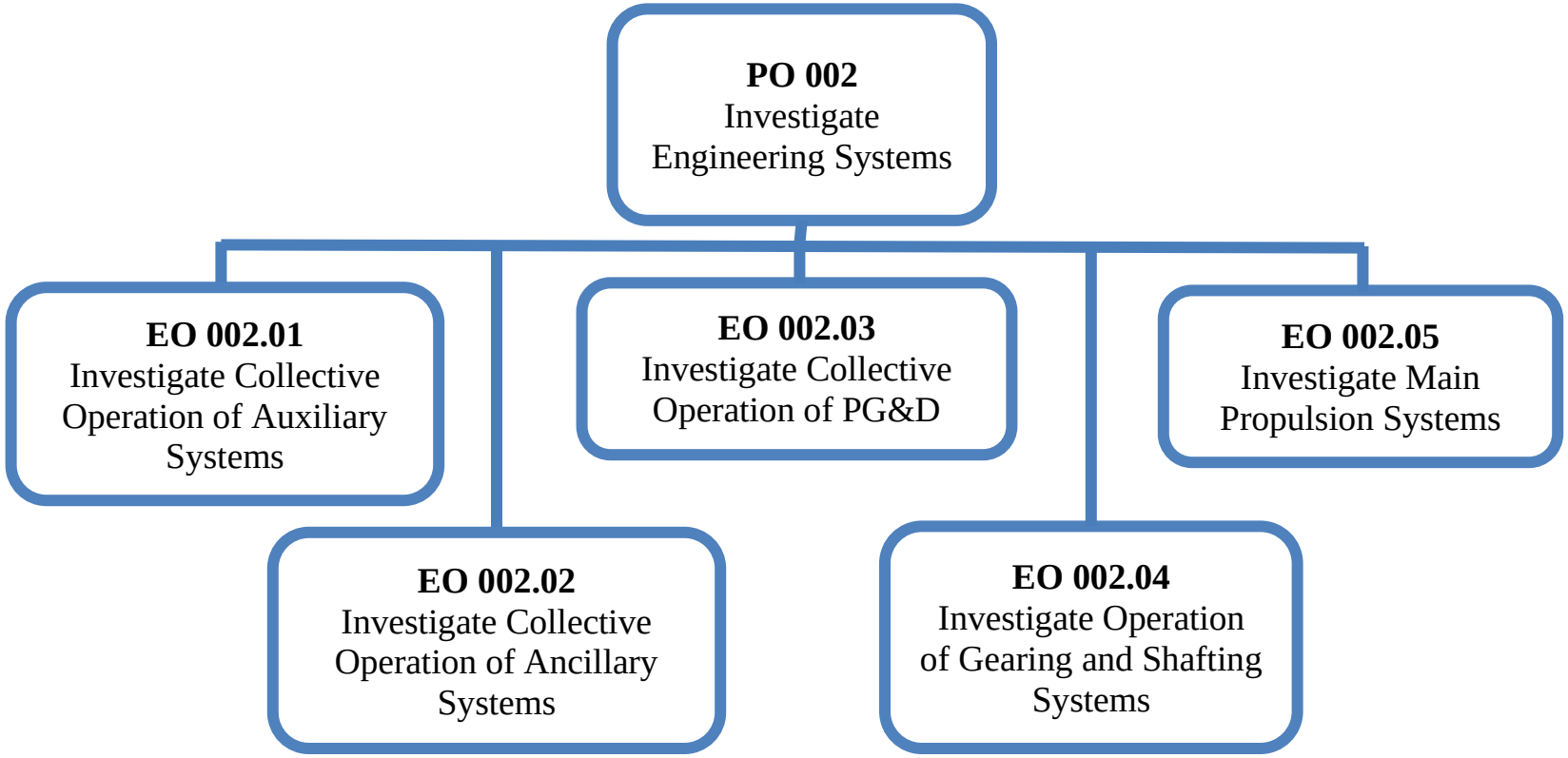
Task No	Task Statement	PO
CT0350	Supervise engineering pre-sailing checks	001
HT0040	Operate IPMS man machinery interface	001
HT0070	Operate IPMS equipment health monitoring	001
HT0050	Operate IPMS software utilities	001
HT0060	Operate IPMS applications	001
ALT1520	Test components with built-in test equipment (BITE)	001
AKT1090	Investigate system operation logs	001
AMT0160	Interpret drawings/schematics	001
IT0380	Manage damage control conditions	001
IT0510	Monitor damage control system	001
ET0230	Lead collective operation of auxiliary systems	001
XT0090	Operate steering system remotely	001
DT0120	Operate fuel oil filling/transfer system remotely	001
ATT0920	Operate fluid piping system	001
WT0090	Monitor ship’s stability	001
WT0040	Assess ship’s stability	001
CT0270	Monitor liquid level management system	001
WT0010	Adjust ship’s stability	001
WT0110	Record ship’s stability variables	001
DT0060	Lead collective operation of ancillary systems	001
DT0070	Operate ancillary systems	001
CT0280	Operate generators in all modes	001
CT0060	Configure ship’s electrical load	001
ABT0050	Lead operation of gearing and shafting systems	001
ABT0170	Operate main gearing and shafting systems	001
BT0200	Testing operation of gearbox	001
ABT0150	Operate gearbox clutch remotely	001
ABT0080	Monitor shaft line operation	001
ABT0070	Monitor gearbox operation remotely	001
RT0010	Operate main propulsion system in all modes	001
ABT0210	Operate propulsion control remotely	001
QT0390	Operate main diesel engine remotely	001
ABT0200	Operate propulsion control locally	001
QT0350	Operate main diesel engine locally	001
GT0010	Advise command on machinery/system status	001
CT0130	Investigate collective operation of auxiliary systems	002
AVT1330	Troubleshoot auxiliary diesel engine internal components	002
AVT1290	Troubleshoot auxiliary diesel engine control system	002
AVT1360	Troubleshoot auxiliary diesel engine sea water cooling system	002
AVT1300	Troubleshoot auxiliary diesel engine exhaust system	002
AVT1350	Troubleshoot auxiliary diesel engine starting system	002

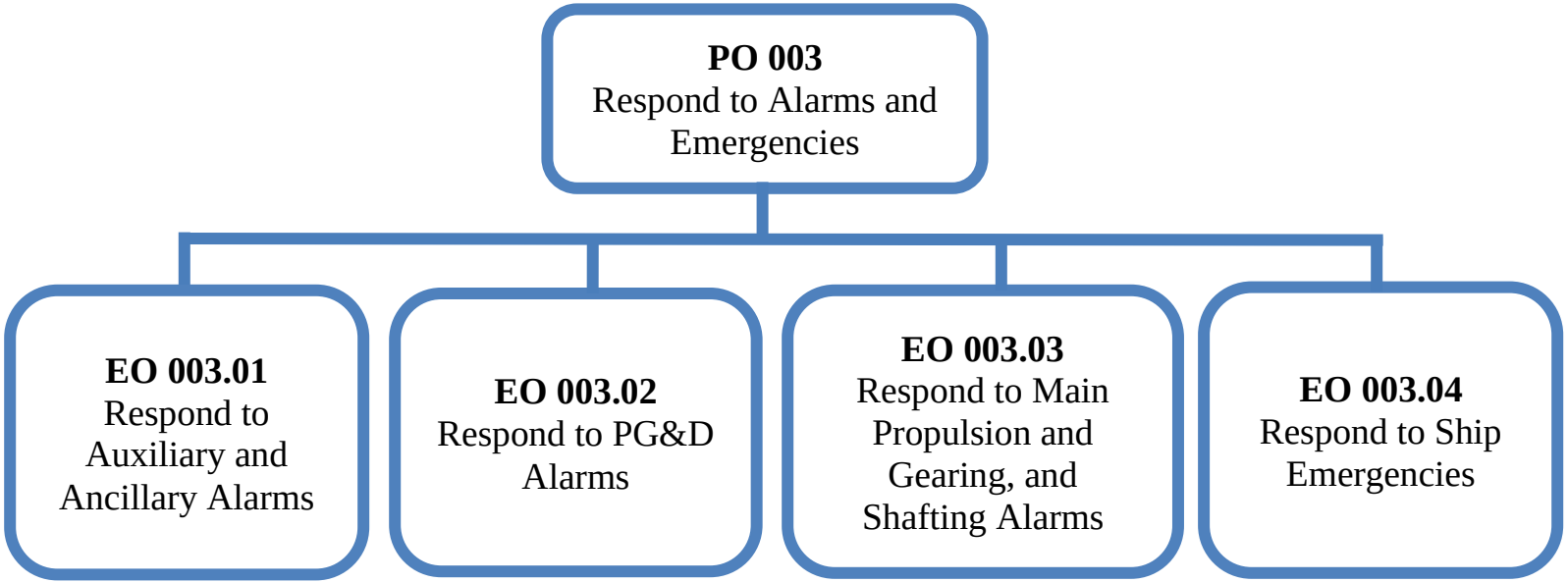
AVT1320	Troubleshoot auxiliary diesel engine fresh water system	002
AVT1310	Troubleshoot auxiliary diesel engine fuel system	002
AVT1340	Troubleshoot auxiliary diesel engine lube oil system	002
AVT1280	Troubleshoot auxiliary diesel engine combustion air system	002
AHT0890	Troubleshoot firemain	002
AHT0840	Troubleshoot bilge system	002
CT0120	Investigate collective operation of ancillary systems	002
AHT0820	Troubleshoot ancillary systems	002
AHT0880	Troubleshoot main seawater circulation system	002
AHT0850	Troubleshoot main lube oil system filling and transfer system	002
CT0150	Investigate collective operation of power generation and distribution system	002
CT0160	Investigate operation of gearing and shafting system	002
BT0050	Evaluate gearbox function	002
BT0060	Evaluate shaftline function	002
CT0140	Investigate collective operation of main propulsion system	002
CT0430	Troubleshoot main propulsion system	002
AHT0860	Troubleshoot fuel oil system	002
JT0040	Advise command on HVAC air distribution system capabilities and limitations	003
JT0030	Advise command on hotel system capabilities and limitations	003
CT0030	Advise command on power generation and distribution system	003
PT0010	Advise command on ship’s power configuration	003
CT0050	Advise command on ship’s electrical load	003
CT0330	Report on power generation and distribution system	003
GT0010	Advise command on machinery/system status	003
CT0040	Advise command on propulsion capabilities and limitations	003
CT0020	Advise command on marine systems operational capabilities and limitations	003
CT0010	Advise command on marine systems engineering parameters	003
CT0340	Report on propulsion deficiencies	003
CT0310	Report on machinery limitations	003
CT0320	Report on marine systems engineering parameters	003
IT0470	Manage propulsion plant operation in emergency stations	003
WT0020	Advise command on ship’s stability	003

ANNEX C – QUALIFICATION STANDARD AND PLAN WRITING BOARD RECORD OF PROCEEDINGS

ANNEX D - SCALARS




PO 002
Investigate Engineering Systems
EO 002.01
Investigate Collective Operation of Auxiliary Systems
EO 002.02
Investigate Collective Operation of Ancillary Systems
EO 002.03
Investigate Collective Operation of PG&D
EO 002.04
Investigate Operation of Gearing and Shafting Systems
EO 002.05
Investigate Main Propulsion Systems


PO 003
Respond to Alarms and Emergencies
EO 003.01
Respond to Auxiliary and Ancillary Alarms
EO 003.02
Respond to PG&D Alarms
EO 003.03
Respond to Main Propulsion and Gearing, and Shafting Alarms
EO 003.04
Respond to Ship Emergencies

ANNEX E – ON JOB PERFORMANCE RECORD (OJPR)

(Certificate of Qualification)

_____

Name and Initials Rank SN

_____

Date Training Started Expected Date of Completion

THE OJPR MUST BE COMPLETED IN ACCORDANCE WITH THE INSTRUCTIONS BELOW. INCORRECT OR INCOMPLETE OJPR WILL BE RETURNED FOR RE-SUBMISSION.

INSTRUCTIONS

1. One copy of this OJPR, hereinafter referred to as the master copy, shall be made up for each individual beginning OJT. It shall be kept and maintained by the supervisor. Recording of the individual's progress shall be entered in Monitor Mass under the Training Tracker. The NFS for the OJPR must maintain Training Tracker fields as outlined in Monitor Mass.

2. As the trainee completes each Performance Objective to the required standard, the MSE Training PO will date and sign the applicable item on the master and duplicate copies of the OJPR sheet and enter progress in the Monitor Mass Training Tracker (MMTT).

3. If it is not possible to conduct OJPR for some POs due to equipment not fitted or not issued (e.g. Work Period), the MSE Training PO is directed to contact Training Manager NPTG HQ for guidance. Extension of training time might be required.

4. See Appendix 1 for OJPR sign off list.

5. Appendix 1: Performance Checks require the trainee to demonstrate the listed competencies as described in Chapter 4. A series of confirmation questions will be asked from each EO to demonstrate the knowledge of the trainee before granting qualification. The trainee will have six (6) months from being posted to an operational ship to complete Appendix 1.

6. On completion of this OJPR, the MSE Training PO shall check box PO 006 in MMTT. This will update automatically Monitor Mass, MPRR and auto generate a message.

7. If you wish to make a change, send your request to: +NPTG MITE@NPTG HQ@Esquimalt

PROGESS REVIEW

11. Progress review shall be carried out monthly by the Divisional Officer and CHOD.

OJPR Reviewed By (Rank, Position, Name)	Signature	Date

TRAINING COMPLETION CERTIFICATE

Date training completed: _____

_____

MAR TECH Chief of Department (CHOD) (Print and sign) Date

CERTIFICATION OF QUALIFICATION

I certify that _____ has completed training to the standards stated in the Marine Technician Engineering Watchkeeper HFX Class Package. The member is qualified to carry out the duties of a Marine Technician Engineering Watchkeeper and is (MITE 121633) qualified.

_____

Engineering Officer (Print and Sign) Date

ON-JOB PERFORMANCE RECORD SHEET

1. The remarks column in Appendix 1 shall be used for the following information only:

   a. The unit in which the training and assessment were carried out; and

   b. Any unusual circumstances which preclude compliance with an objective.

APPENDIX 1

PO/EO	Performance Statement	Signature / Comments
002.01	The Trainee shall Conduct Watchkeeping operations to the satisfaction of the MSEO by::	
002.01.001	Conduct pre-watch and on-watch rounds.	
002.01.002	Conduct flash-up and shut-down routine.	
002.01.003	Operate propulsion system.	
002.01.005	Transferring fuel.	
002.01.006	Operating propulsion systems locally.	
002.04	The Trainee shall operate the PG&D to the satisfaction of the MSEO by:	
002.04.001	Changing over electrical generators at sea.	
002.05	The Trainee shall operate main gearing to the satisfaction of the MSEO by:	
002.05.001	Operating the gear box locally.	
002.06	The Trainee shall operate the Main Propulsion Engines to the satisfaction of the MSEO by:	
002.06.001	Flash up/shut down MEs locally.	
002.06.002	Flash up/shut down MEs remotely.	

004	The Trainee shall correctly perform delaying drills as Watchkeeper as per reference ORCA Class EOPs. All drills listed in the manual require a signature. If a drill is added/removed from the manual it is to be automatically added/removed from this OJPR.
	All the drills below will be assessed by the Orca Engineering Watchkeeper Instructor.

Loss of both shaft lines	
Loss of one shaft line	
Loss of station in control	
Loss of salt water cooling	
Propulsion engine high freshwater temperature	
Propulsion engine low lube oil pressure	
Propulsion engine inlet air restriction	
Propulsion engine high exhaust temperature	
Propulsion engine overspeed condition	
Propulsion engine high crankcase pressure	
Gearbox low lube oil pressure	

Gearbox high lube oil temperature	
Power failure	
Steering gear breakdown	
RTU #1 failure	
RTU #2 failure	
log strike	
Respond to Fire	
Respond to Flood	
Respond to Man Over Board	
Respond to Fuel Spill	
Respond to HAZMAT Spill	
Final set of drills containing no less than three continuous drills conducted by and to the satisfaction of the MSEO.	
005	The Trainee shall correctly perform the duties of the fueling IC IAW ORCA SEMS without error

Conduct Fueling Safety Brief IAW ORCA SEMS and EOTIs	
Perform Duties as Fueling IC	
Complete Oral Board as Per Appendix 2	

Note: Signatures required for all lines not greyed.

APPENDIX 2

ENGINEERING WATCHKEEPER

QUALIFICATION BOARD ASSESSMENT GUIDE

ORAL EXAMINATION BOARD PROCEDURE

1. The Oral Examination Board conducted at the end of the Engineering Watchkeeper training package will measure the candidate's understanding of the material studied and the engineering systems they have learned to this point. In order to pass the board, the candidate must receive a mark of at least 60% with the first two questions also having a Pass/Fail criteria for the board (i.e. a failure of either results in a failure of the board). Board length will be set at a maximum of two (2) hours. Board dates and times will be coordinated by the ship's MSEO. The board shall be composed of the Ship's MSEO (as board Chair), the Chief of the MSE Department, a PO2/PO1 Mechanical MAR TECH, and a PO2/PO1 Electrical MAR TECH.

2. During the board the candidate shall not have access to any references, and will be supplied with the drawing (supplied by NTDC(P)). The board shall consist of four (4) questions. The first question will be a description of the system drawing. The candidate will be informed which diagram they are required to present 30 minutes prior to the board to allow the candidate to prepare. The second question will be based on Safety and Environmental Policy / Considerations. The board will then take a break to determine if the candidate has successfully challenged the first two questions prior to carrying on with the rest of the board. The remaining questions shall consist of a general stability/watchkeeping question, and a question based on a watchkeeping scenario. The questions will be prepared from the question bank provided by NTDC(P).

3. In the event of a board failure or training difficulties, refer to the MAR TECH Watchkeeper QSP.

4. <u>Board Format</u> (suggested timings for questions noted below)

Q1: Engineering drawing – describe a provided sketch (P/F)
Q2: Safety and Environmental Policy / Considerations
Q3: General Technical knowledge, specifically (one of): a. Stability and liquid handling b. General Watchkeeping

Q4: Scenario based watchkeeper problem (candidate will have time to put plan together)

1. ENGINEERING DRAWING (drawing supplied) (45 min)

The candidate is provided with a drawing 30 minutes before the Board. The drawing is of a system with parameters/other details removed. The drawings to be used for this question are the following:

a. Sea Water Cooling;

b. Compressed Air;

c. Fire main;

d. Shaft line;

e. Switch Board;

f. Steering;

g. Fuel Oil Service and Transfer;

h. Blackwater;

i. Clean Bilge; and

j. Co2 Fire Suppression.

The examination board members will have a copy of the drawing with all relevant information provided. The candidate must explain the following to the Examining Board:

a. System Description to include:

(1) Overview of system; and

(2) Purpose of system.

b. System Operation to include:

(1) Normal seagoing configuration; and

(2) Special operating states.

c. Component Description:

(1) Brief purpose and function of each component from the perspective of the Engineering Watchkeeper (not to be treated as a legacy technician board).

d. Parameters

(1) Normal operating ranges; and

(2) Max/min values and warning/alarm points.

e. Direction of flow for:

(1) Fluids; and

(2) Energy, rotation or other mechanisms, as appropriate.

f. Abnormal operation:

(1) Alternate modes in response to a failure or breakdown and their limitations; and

(2) Considerations to the engineering plant in response to any abnormal mode.

g. Emergency Operation

2. SAFETY AND ENVIRONMENTAL POLLICY / CONSIDERATIONS (15 min)

The candidate will be given a scenario with a safety/environmental concern while on watch. The candidate will describe the safety/environmental actions to take including immediate actions, briefing the watch, advising command and references.

3. GENERAL TECHNICAL (Stability and liquid handling or General Watchkeeping) (30 min)

a. STABILITY AND LIQUID HANDLING

The candidate will be given a very basic stability question (ie changes to fuel schedule, difference in fresh water tank levels), to identify the apparent problems this may cause. This question is intended to ensure the watchkeeper is faced with potential changes to stability that can happen during a watch. Follow on aspects can include providing general recommendations and when to call for assistance in correcting the ship's stability.

b. GENERAL WATCHKEEPING

The candidate will be given a basic watchkeeping question (e.g. rounds to be conducted, flash-up procedure, etc), to identify the actions the watchkeeper should take. This question is intended to determine the watchkeeper's general knowledge. This may include general recommendations and when to call for assistance.

SCENARIO BASED (30 min)

The candidate will be given an engineering watchkeeping scenario and will be asked to explain how they would react. The scenario could be taken from any type of watchkeeping incident such as a failure of an engine to start, flash-up procedures or emergencies. Reactions to the scenario are to include:

a. watchkeeper actions;

b. listing personnel/technical resources available to assist; and

c. briefs to Senior Watchkeeper/CHOD/MSEO.

The candidate will be given time to compile their responses prior to responding to the board members.

Copy _____ of 4

Engineering Watchkeeper – Qualification Board Questions

Candidate:	Ship:	Date:
MSEO:	Chief of Department:	PO2/PO1 MAR TECH Mechanical
PO2/PO1 MAR TECH Electrical		

INSTRUCTIONS

Questions are scored out of 10 pts; 0-8 pts for knowledge / understanding and 0-2 pts for presentation / conduct. A minimum of 60% is required for each of the first two questions in order for the board to carry on past the second question.

KNOWLEDGE / UNDERSTANDING:

0 No knowledge. Incapable of answering main part of the question, including where the system/equipment is located, what it looks like or how to begin even researching the topic.

1 Extremely low knowledge. Incapable of answering main part of the question other than where the system/equipment is located, what it looks like or knows where/who to go to in order to obtain more information.

2 Very low knowledge. Continuous assistance was required to answer the entire main question and answers were insufficient in detail.

3 Low knowledge. Continuous assistance was required to answer half or fewer of the main question and answers were insufficient in detail.

4 Marginal knowledge. Frequent assistance was required to answer the main question in acceptable detail. Frequent assistance was required to answer follow up questions and answers were insufficient in detail.

5 Adequate knowledge. Minimal assistance was required to answer the main question in acceptable detail. Frequent assistance was required to answer follow up questions and answers were insufficient in detail.

6 Proficient knowledge. Minimal assistance was required to answer the main question and follow up questions in acceptable detail.

7 Strong knowledge. No assistance was required to answer the main question and follow-up questions in acceptable detail.

8 Exemplary knowledge. Answered main question completely and elaborated in detail on specific aspects such that follow-up questions were unnecessary.

PRESENTATION / CONDUCT:

0 Displayed no confidence or professionalism with board members. Answers/diagrams were vague, unorganized and non-committal.

0.5 Displayed poor confidence or professionalism with board members. Answers/diagrams were partially vague and partially unorganized.

1 Displayed confidence and professionalism with board members. Answers/diagrams were organized.

1.5 Displayed strong confidence and professionalism with board members. Answers/diagrams were organized well.

2 Displayed exemplary confidence and professionalism with board members. Answers/diagrams were highly organized.

Engineering Watchkeeper

QUALIFICATION BOARD ASSESSMENT SHEET

Candidate:	Ship:	Date:

NOTE: Intent is for each assessor to ask one question per section.

Question		Content (0-8)	Conduct (0-2)	Score (S) (0-10)	Comments	
System Drawing	1					
Safety Environmental policy / considerations	2					
General Technical (Stability or Liquid handling, General Watchkeeping, Administrative/ leadership)	3					
Watchkeeping Scenario	4					
					Total =	
					RESULT (circle one) =	PASS / FAIL

Board Member Comments:

Board Member ID (Mandatory):

Name: _____ Signature:_____

BOARD Average Score

AVERAGE: [(1)___ + (2)___ + (3)___ + (4)___] / 4 =

MAR TECH Engineering Watchkeeper Qualification Board Checklist

Candidate: _____ Board date: _____

Oral Board Prep:

a. OJPR complete and reviewed
b. Board questions prepared
c. Score sheets prepared
d. Watchkeeping Qualification prepared

Post Board Action:

a. Score sheets signed, corrected and averaged
b. Watchkeeping Qualification signed if member passed
c. Successful completion message drafted and released

COMPOSITION OF MAR TECH ENGINEERING WATCHKEEPER BOARD

Each board exam is to include 8 questions, of which only 4 are used, one per type. Where multiple questions are provided, the board may decide which to use prior to the board start.
The question categories are laid out in the chart below.
A board composition shall be IAW this QSP.
The first question will be asked by the Board Chair.
The second question will be asked by the Chief of Department.
The remaining questions will be asked by the other board members to be determined by the Board Chairperson beforehand.

Question types	Quantity
Engineering problem - drawing	1
Safety Environmental policy / considerations	2
General Technical (Stability & Liquid Handling OR General Watchkeeping)	2 (1 stabi, 1 general)
Watchkeeper Scenario	3

ANNEX F – ACRONYMS AND ABBREVIATIONS

Acronym and Abbreviation	Term or Definition
CBRN	Chemical Biological Radiological Nuclear
DG	Diesel Generator
EHM	Electronic Health Monitor
GLM	General Load Monitor
GT	Gas Turbine
HVAC	Heating Ventilation Air Conditioning
IMO	International Maritime Organization
IPMS	Integrated Platform Monitoring System
LCC	Local Control Console
LCP	Local Control Panel
LLMS	Liquid Level Management System
LOP	Local Operating Panel
MCC	Machinery Control Console
MDE	Main Diesel Engine
NEM	Naval Engineering Manual
PCOT	Propulsion Control Operator Trainer
PT	Power Turbine
PTIT	Power Turbine Inlet Temperature
SSS	Synchronous Self Shifting

ANNEX G - SUGGESTED SEQUENCES AND TIMINGS

1. The ORCA Engineering Watchkeeper course is broken into five POs, each of which contain EOs related to each other for ease of training and subject material. The suggested sequence for delivering the training is as follows with the OJPR completed last:

   a. POs 1 through 3 must be completed in numerical order as follows:

PO 001: Explain the operation of engineering systems					
	EO	**EC**	**PC**	**Multiplier**	
EO 001.01	175	50	200	1	
EO 001.02	1175	100			
EO 001.03	275	50			
EO 001.04	100	25			
EO 001.05	125	25			
EO 001.06	275	50			
Subtotals:	2125	300	200		
Delivery Method:		**F2F**	**Total (Mins):**		2625
			Total (Days):		6.75

PO 002: Operate IPCS console					
	EO	**EC**	**PC**	**Multiplier**	
EO 002.01	225	50	200	1	
EO 002.02	325	50			
EO 002.03	100	50			
EO 002.04	150	50			
EO 002.05	150	50			
EO 002.06	100	50			
EO 002.07	300	50			
Subtotals:	1350	350	200		
Delivery Method:		**F2F**	**Total (Mins):**		1900
			Total (Days):		4.75

PO 003: Investigate engineering systems				
	EO	**EC**	**PC**	**Multiplier**
EO 003.01	450	50	OJPR	1
EO 003.02	100	50		
EO 003.03	100	50		
EO 003.04	100	50		
EO 003.05	325	50		
Subtotals:	1175	250		
Delivery Method:		**F2F**	**Total (Mins):**	1625
			Total (Days):	3.29

PO 004: Respond to Alarms and Emergencies				
	EO	**EC**	**PC**	**Multiplier**
EO 004.01	500	50	OJPR	8
EO 004.02	175	50		
EO 004.03	300	50		
EO 004.04	150	50		
Subtotals:	1125	200	0	
Delivery Method:		**F2F**	**Total (Mins):**	1325
			Total (Days):	3.35

PO 005: Perform the duties of Fuelling I/C				
	EO	**EC**	**PC**	**Multiplier**
EO 004.01	100	50	OJPR	8
EO 004.02	75	50		
Subtotals:	175	100		
Delivery Method:		**F2F**	**Total (Mins):**	275
			Total (Days):	0.7

Note: The timings presented above do not contain any accounting of TAT, BAT or PT. The column labeled as “multiplier” indicates the number of times that testing runs must be conducted to capture all 8 students once.

2. The following table provides the expected course duration information with the maximum trainees considered (12):

	EO	**EC**	**PC**	**TAT**	**BAT**	**PT**	**TTT**	**Divisions**
Minutes	6590	750	6700	200	400	800	200	000
Days	12.09	7.5	16.75	.5	1	2	.5	0

Total	**Time**
Days	40
Weeks	8
Months	2

3. The TE is expected to review all ECs and PCs to confirm that they can accomplish the required testing within the allocated times. If the TE cannot achieve the aim, two options are to be considered: purchasing required training material to meet the timings or increase the time allocated to the course to achieve the aim.

ANNEX H - TRAINING RESOURCES

1. The following is a list of the resources identified as being required to conduct the Marine Technician ORCA Engineering Watchkeeper Course. Some of the resources listed currently exist at the Naval Fleet Schools; however, there are new resources that will be required to maximize the efficiency of training delivery and testing. Purchasing the new items will increase the quality of the student graduating from the training, and significantly reduce the number of required training days ultimately saving time and money over the long-term.

2. The following is a list of classroom requirements:

   a. One large classroom to accommodate maximum number of candidates which include:

      (1) Individual candidate's desks c/w chair;

      (2) Smart Board;

      (3) Dry erase board;

      (4) Projector (power point capable) & screen;

      (5) BluRay Player & TV;

      (6) Podium; and

      (7) Student handouts and references.

   b. Operational ORCA Class Vessel to accommodate maximum number of trainees.

      (1) Current course loading capacity is 8 trainees.

ANNEX I – COSTING

	UNIT	PRICE	QTY	TOTAL
Supplies (STUDENTS)				
1" Binder	EA	$9.39	12	$112.68
Eraser	EA	$2.15	12	$25.8
Blue Pen	EA	$0.69	12	$8.28
Pencil	BOX	$6.39	1	$6.39
Lined Paper	PKG	$9.99	12	$119.88
Highlighter	BOX	$21.48	1	$21.48
SUBTOTAL				$294.51
Supplies (INSTRUCTOR)				
Blue Pen	EA	$0.69	2	$1.38
Red Pen	EA	$0.69	2	$1.38
Pencils	BOX	$6.39	1	$6.39
Post It Notes	PKG	$5.89	4	$23.56
Writing Pad 8.5" x 11"	EA	$4.79	4	$19.16
Whiteboard Markers	PKG	$6.99	1	$6.99
Whiteboard Eraser	EA	$11.99	1	$11.99
Paper Clips	BOX	$1.09	4	$4.36
Sharpie	EA	$3.69	2	$7.38
CDROM	PKG	$18.33	1	$18.33
SUBTOTAL				$82.59
*This table does not include costing for travel and R&Q				

ANNEX J- COURSE HISTORY

The USQ ORCA Engineering Watchkeeper QSP was developed from the legacy QSP material and input from representatives from ER, ET, and HT legacy MOSIDs. This course is a follow-on from the Marine Technician Machinery Maintainer S2 (AB) USQ or the S3 (LS) Technician Course, and is a prerequisite for employment as an Engineering Watchkeeper on board ORCA Class Vessels.

February 2021 – Minor formatting changes. Prerequisites for loading of course were updated.

25 October 2022 – Amendment Chapter 1 para 7.b. Email between CPO2 Sutton Mar Tech TM and CPO2 Gourlay, NavRes DNP Mar Tech OM. Added line v.

ANNEX K – ASSESSMENT PLAN

EC/PC	Scope Covered	Purpose	How	Pass/Fail Criteria	Impact
EC001.01		Explain IPCS/BCC	Homework	Nil	Trainee issued Trainee Experiencing Difficulty (TED) and provided additional training/instruction.
EC 001.02		Explain Ancillary, Auxiliary and Domestic Systems	Homework	Nil	Trainee issued Trainee Experiencing Difficulty (TED) and provided additional training/instruction.
EC 001.03		Explain PG&D	Homework	Nil	Trainee issued Trainee Experiencing Difficulty (TED) and provided additional training/instruction.
PC 001.03		Assessment EC 001.01 through EC 001.03	Written test	60%	First PC Fail results in a retest Second PC failure results in a TRB IAW NPTORD. Remedial training/counselling as required.
EC 001.04		Explain Shafline Arrangements	Homework	Nil	Trainee issued Trainee Experiencing Difficulty (TED) and provided additional training/instruction.
EC 001.05		Explain ZF Gearbox	Homework	Nil	Trainee issued Trainee Experiencing Difficulty (TED) and provided additional training/instruction.
EC 001.06		Explain ME	Homework	Nil	Trainee issued Trainee Experiencing Difficulty (TED) and provided additional training/instruction.

PC 001.06	Assessment of EC 001.04 through EC 001.07	Written test	60%	First PC Fail results in a retest Second PC failure results in a TRB IAW NPTORD. Remedial training/counselling as required.
EC 002.01	To assess the trainees ability to operate bridge control console/IPCS.	Developmental Observations	Nil	Trainee issued Trainee Experiencing Difficulty (TED) and provided additional training/instruction.
EC 002.02	To assess the trainees ability to operate auxiliary systems.	Developmental Observations	Nil	Trainee issued Trainee Experiencing Difficulty (TED) and provided additional training/instruction.
EC 002.03	To assess the trainees ability to lead collective operation of ancillary systems.	Developmental Observations	Nil	Trainee issued Trainee Experiencing Difficulty (TED) and provided additional training/instruction.
EC 002.04	To assess the trainees ability to operate Domestic Systems.	Developmental Observations	Nil	Trainee issued Trainee Experiencing Difficulty (TED) and provided additional training/instruction.
EC 002.05	To assess the trainees ability to operate Power Generation and Distribution systems.	Developmental Observations	Nil	Trainee issued Trainee Experiencing Difficulty (TED) and provided additional training/instruction.
EO 002.06	To assess the trainees ability to operate gear box and shaft line.	Developmental Observations at Sea	Nil	Trainee issued Trainee Experiencing Difficulty (TED) and provided additional training/instruction.
EO 002.07	To assess the trainees ability to operate the Main Propulsion System locally and Remotely.	Developmental Observations At Sea	Nil	Trainee issued Trainee Experiencing Difficulty (TED) and provided additional training/instruction.

PC 002 Part A and B	EO 002.01 – 002.07	To assess the trainee's ability to Bridge Control Consoles.	Practical Assessment at Sea	no critical failures as per PC checklist.	First PC Fail results in a retest Second PC failure results in a TRB IAW NPTORD. Remedial training/counselling as required.
EC 004.01	EO 003.01 – 003.02, 004.01	To assess the trainees ability to respond to auxiliary, ancillary and domestic alarms.	Practical Assessment at Sea		Trainee issued Trainee Experiencing Difficulty (TED) and provided additional training/instruction
EC 004.02	EO 003.03 004.02	To assess the trainees ability to respond to PG & D alarms.	Practical Assessment at Sea	TBD	Trainee issued Trainee Experiencing Difficulty (TED) and provided additional training/instruction.
EC 004.04	EO 004.03 and 004.04	To assess the trainee's ability to respond to ship emergencies.	Practical Assessment at Sea.		Trainee issued Trainee Experiencing Difficulty (TED) and provided additional training/instruction.
EC 005.02	EO 005.01 and 005.02	To assess the trainee's ability to assume the duties of Fueling IC.	Practical Assessment at Sea.		Trainee issued Trainee Experiencing Difficulty (TED) and provided additional training/instruction.

OJPR	PO 002 003, 004 and 005	To assess the trainee's ability to investigate engineering systems and respond to alarms and emergencies.	Practical Assessment with final oral board.	Completion of OJPR package.	First Board Fail results in a second board with a training extension Second Board failure results in a TRB IAW NPTORD. Remedial training/counselling as required.

ANNEX L – TECHNOLOGY ENABLED LEARNING

Nil